Understanding the Classical Orthogonal Polynomials

Jacobi, Laguerre, Hermite, and the associated families

A self-contained first course

November 2, 2026

Scope of these notes

These notes are a self-contained first course on the classical orthogonal polynomials: the Jacobi, Laguerre and Hermite families, together with the fourth family of Bochner's classification, the Bessel polynomials, which is classical as a differential equation but is not orthogonal on the real line. They assume one semester of calculus and one semester of linear algebra, and nothing else. Every object used is defined in §1, and no statement is quoted without either a proof or a numerical check that the reader can repeat.

The first half of the note puts the three families side by side in one normalisation and derives, once and for all, the facts that are usually quoted family by family: the Pearson equation, the Rodrigues formula, the differential equation, the three-term recurrence in both the standard and the monic normalisation, the norm, the structure relation and the identity $h_n = a_n^2 h_{n-1}$. The value of doing this in one place is that the three families are then visibly the same object with different parameters, and the limits between them become computations rather than assertions.

The second half does something the first half does not: it takes the three-term recurrence and *shifts its coefficients by a parameter* c. The coefficient b_n is replaced by b_{n+c} and a_n^2 by a_{n+c}^2, and the resulting sequence of polynomials is called the *associated* family of order c. For $c = 0$ nothing changes; for $c = 1, 2, \dots$ the new family is still orthogonal, but it is no longer Jacobi, Laguerre or Hermite, and its weight is no longer a power times an exponential. The central result of the second half is that the new weight can nevertheless be written in closed form:

$$w_c(x) = \frac{w_{c-1}(x)}{a_c^2 \, |F_{c-1}(x + \mathrm{i}0)|^2},$$

where F_{c-1} is the Stieltjes transform of the associated mass distribution one step down. For the three classical families the transforms are elementary, so the weights are too: we obtain $6/(\pi^2 + \ln^2 \frac{1+x}{1-x})$ for the associated Legendre family, $e^x/(\mathrm{Ei}(x)^2 + \pi^2)$ for the associated Laguerre family at $\alpha = 0$, and $2e^{x^2}/(\pi^{3/2} \, |\mathrm{erfc}(-\mathrm{i}x)|^2)$ for the associated Hermite family. Each of these weights, and the orthogonality of the corresponding polynomials, is checked numerically in the note.

What is assumed. Calculus (integration, integration by parts, Taylor expansion, separable differential equations) and linear algebra (matrices, determinants, eigenvalues). Factorials, binomial coefficients, the Pochhammer symbol, the gamma function and the beta function are defined in §1.5, and the few facts from calculus, complex arithmetic and linear algebra that are used repeatedly are collected in

§1. No measure theory is assumed: mass distributions are introduced concretely in §6 as weights and finite sums of point masses, and every statement about them is either proved by calculus or verified numerically.

Conventions. The variable is x unless the family forces otherwise. Q_n denotes the *monic* member of a family, $Q_n(x) = x^n + \dots$, and P_n, $L_n^{(\alpha)}$, H_n the standard normalisations of Jacobi, Laguerre and Hermite. The recurrence is written

$$xQ_n = Q_{n+1} + b_n Q_n + a_n^2 Q_{n-1}, \qquad Q_{-1} = 0, \quad Q_0 = 1, \tag{1}$$

so that b_n is real and $a_n^2 > 0$ in the positive-definite case. Parameters are real and in the range stated for each family. The empty product is 1 and the empty sum is 0.

On the numerical checks. The symbolic identities (recurrences, structure relations, differential equations, and the physics reductions of §4) were checked with Maxima 5.49 in exact or symbolic arithmetic, with every residual identically zero. The limit tables of §5 were computed from the exact recurrence coefficients in 30-digit arithmetic; quadratures were run in double precision. The Gram-matrix checks of §7 and the residual of the convergent identity (142) were also computed in double precision and are quoted only to the accuracy that precision supports. Where a formula is quoted from the literature rather than derived, it is one of the standard formulas of Chapter 18 of the DLMF [1], and it is also checked against the derived formulas.

Roadmap. Section 1 is the background: inner products, orthogonality, the Gram–Schmidt construction, the three-term recurrence, and the notation and facts from calculus, complex numbers and linear algebra that the rest of the note uses. Section 2 explains what "classical" means and proves Bochner's theorem, the characterisation that produces the four families. Section 3 studies Jacobi, Laguerre and Hermite in detail. Section 4 shows where they come from in physics: Laplace's equation in spherical coordinates, the multipole expansion, the hydrogen atom and the harmonic oscillator. Section 5 then shows how the three families are limits of one another. Section 6 passes from polynomials to matrices: the Jacobi matrix, the Stieltjes transform, the continued fraction, and the second-kind polynomials. Section 7 is the associated case, and §8–§9 treat zeros, the numerical checks and the usual confusions. The exercises at the end are part of the text: several of them fill in computations that would interrupt the flow.

Further reading. The classical references are [3] and [4]; [5] develops the hypergeometric theory, [7] is the standard compendium of formulas, and [6] treats the computational side. The formulas of Chapter 18 of the DLMF [1] are the ones used for cross-checks throughout.

Classical in one page

Bochner's theorem says that a family of orthogonal polynomials can satisfy a second-order differential equation with polynomial coefficients only in four ways. The data are as follows.

	Jacobi	Laguerre	Hermite	Bessel
interval	$(-1,1)$	$(0,\infty)$	$\mathbb{R}$	unit circle
weight w	$(1-x)^\alpha(1+x)^\beta$	$x^\alpha e^{-x}$	e^{-x^2}	see §2
σ	$1-x^2$	x	1	x^2
τ	$\beta-\alpha-(\alpha+\beta+2)x$	$\alpha+1-x$	$-2x$	$ax+b$
λ_n	$n(n+\alpha+\beta+1)$	n	$2n$	$-n(n+a-1)$
parameters	$\alpha,\beta>-1$	$\alpha>-1$	—	$b\neq 0$

Table 1: The four families of Bochner's theorem. The first three are orthogonal on the real line with a positive weight; the fourth is not.

The differential equation, the weight and the Rodrigues formula are linked by

$$\sigma(x)p_n''(x)+\tau(x)p_n'(x)+\lambda_n p_n(x)=0, \qquad (\sigma w)'=\tau w, \qquad p_n \propto \frac{1}{w}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\big(\sigma^n w\big). \tag{2}$$

In the monic normalisation the recurrence coefficients and the norms are

	Jacobi	Laguerre	Hermite
b_n	$\dfrac{\beta^2-\alpha^2}{(2n+\alpha+\beta)(2n+\alpha+\beta+2)}$	$2n+\alpha+1$	0
a_n^2	$\dfrac{4n(n+\alpha)(n+\beta)(n+\alpha+\beta)}{(2n+\alpha+\beta)^2(2n+\alpha+\beta+1)(2n+\alpha+\beta-1)}$	$n(n+\alpha)$	$\dfrac{n}{2}$
k_n	$\dfrac{(n+\alpha+\beta+1)_n}{2^n\,n!}$	$\dfrac{(-1)^n}{n!}$	2^n
h_n^{mon}	h_n^{std}/k_n^2	$n!\,\Gamma(n+\alpha+1)$	$\dfrac{n!\sqrt{\pi}}{2^n}$

Table 2: Monic recurrence coefficients, leading coefficients and monic norms. h_n^{std} is the norm in the standard normalisation and $h_n^{\mathrm{mon}}=h_n^{\mathrm{std}}/k_n^2$.

The identities that organise the whole subject are

$$xQ_n = Q_{n+1}+b_nQ_n+a_n^2Q_{n-1}, \tag{3}$$

$$h_n = a_n^2 h_{n-1}, \tag{4}$$

$$Q_n'(x;\text{parameters}) = n\,Q_{n-1}(x;\text{parameters}+1), \tag{5}$$

$$F_0(z)=\int\frac{\mathrm{d}\nu(t)}{z-t}=\cfrac{1}{z-b_0-\cfrac{a_1^2}{z-b_1-\cfrac{a_2^2}{z-b_2-\cdots}}}, \tag{6}$$

$$\nu_c'(x)=\frac{\nu_{c-1}'(x)}{a_c^2\,|F_{c-1}(x+\mathrm{i}0)|^2}, \tag{7}$$

where in (5) "parameters+1" means $(\alpha, \beta) \to (\alpha+1, \beta+1)$ for Jacobi, $\alpha \to \alpha+1$ for Laguerre, and nothing for Hermite; ν in (6) is the normalised weight $w\,\mathrm{d}x/h_0$; and (7) is the associated-weight formula of §7. The rest of the note is the derivation and the explanation of these five lines, in the order in which the objects appear.

1 Background

This section collects, in one place, everything the rest of the note uses. A reader who knows the material can skip to §2; a reader who does not should read it once, because the same handful of facts are used on almost every page.

1.1 Polynomials and the monic normalisation

A *polynomial* in x is a finite sum $c_0 + c_1 x + \cdots + c_n x^n$ with numerical coefficients. Its *degree* is the largest j with $c_j \neq 0$, and c_j is the *leading coefficient*. A polynomial of degree n is *monic* if its leading coefficient is 1, that is, if it has the form $x^n + (\text{lower powers})$.

Two facts are used constantly. First, a polynomial of degree n is determined by its $n+1$ coefficients, so a statement about a family of polynomials is a statement about finitely many numbers for each n. Second, the polynomials $1, x, x^2, \ldots, x^n$ form a basis of the space of polynomials of degree at most n: every such polynomial is a unique linear combination of them. Equivalently, every polynomial of degree at most n is a unique linear combination of any $n+1$ polynomials $Q_0, \ldots, Q_n$ with $\deg Q_k = k$, and in that combination the coefficient of Q_n is the ratio of the two leading coefficients. This is the fact behind the expansion used in the proof of the three-term recurrence below.

We work with the *monic* normalisation because it removes the arbitrary constant: every family of orthogonal polynomials can be made monic in exactly one way, by dividing each member by its leading coefficient. The standard normalisations used in tables are recovered by multiplying back; the passage between the two is described after the three-term recurrence and used in §3.

1.2 Weights, inner products and orthogonality

A *weight* on an interval I of the real line is a function $w \geq 0$ with $\int_I w(x)\,\mathrm{d}x$ finite and positive. The interval may be finite like $(-1, 1)$, half-infinite like $(0, \infty)$, or the whole line. Given a weight, define the *inner product*

$$\langle f, g \rangle = \int_I f(x) g(x) w(x)\,\mathrm{d}x. \tag{8}$$

It is symmetric, $\langle f, g\rangle = \langle g, f\rangle$; it is linear in each argument, $\langle af + bg, h\rangle = a\langle f, h\rangle + b\langle g, h\rangle$; and it is positive definite, $\langle f, f\rangle > 0$ for every f that does not vanish identically (for a continuous f on the interval). The *norm* of f is $\|f\| = \sqrt{\langle f, f\rangle}$. Two functions are *orthogonal* if $\langle f, g\rangle = 0$; a family is *orthonormal* if in addition every member has norm one.

The point of the definition is that it makes functions behave like vectors: $\langle f, g\rangle$ is the analogue of the dot product, and $\|f\|$ the analogue of length. Orthogonality means the same thing in both settings, namely that the "components" do not overlap; this is why the geometry of $\mathbb{R}^n$ can be used below without any new idea.

Example 1. *On $I=(-1,1)$ with $w=1$, every odd power integrates to zero over the symmetric interval, so $\langle 1,x\rangle=0$ and $\langle x,x^2\rangle=0$. The nonzero ones are $\langle 1,1\rangle=2$, $\langle x,x\rangle=\frac{2}{3}$ and $\langle x^2,x^2\rangle=\frac{2}{5}$.*

The numbers $m_k=\int_I x^k w\,\mathrm{d}x$ are the *moments* of w; we assume throughout that all moments are finite, which is the case for the three families of this note. The norm of the monic polynomial of degree n is denoted

$$h_n=\langle Q_n,Q_n\rangle=\int_I Q_n(x)^2 w(x)\,\mathrm{d}x>0. \tag{9}$$

1.3 Existence: the Gram–Schmidt construction

Orthogonal polynomials exist for any weight, and the proof is the familiar orthogonalisation procedure from linear algebra.

Proposition 2 (Gram–Schmidt)**.** *Let w be a weight with finite moments. For each $n\geq 0$ there is exactly one monic polynomial Q_n of degree n with $\langle Q_n,p\rangle=0$ for every polynomial p of degree less than n.*

Proof. Set $Q_0=1$. Suppose $Q_0,\dots,Q_{n-1}$ have been constructed and are orthogonal to one another. Put

$$Q_n(x)=x^n-\sum_{k=0}^{n-1}\frac{\langle x^n,Q_k\rangle}{h_k}\,Q_k(x). \tag{10}$$

The right-hand side is monic of degree n, since the subtracted terms have degree at most $n-1$. For $j<n$,

$$\langle Q_n,Q_j\rangle=\langle x^n,Q_j\rangle-\sum_{k=0}^{n-1}\frac{\langle x^n,Q_k\rangle}{h_k}\langle Q_k,Q_j\rangle=\langle x^n,Q_j\rangle-\frac{\langle x^n,Q_j\rangle}{h_j}h_j=0,$$

because $\langle Q_k,Q_j\rangle=0$ for $k\neq j$ and $\langle Q_j,Q_j\rangle=h_j$. Hence Q_n is orthogonal to $Q_0,\dots,Q_{n-1}$, and therefore, by linearity, to every polynomial of degree less than n. Uniqueness: if Q and $\widetilde{Q}$ were two such monic polynomials, then $Q-\widetilde{Q}$ would have degree less than n and would be orthogonal to itself, so $\langle Q-\widetilde{Q},Q-\widetilde{Q}\rangle=0$ and $Q=\widetilde{Q}$. □

Equation (10) is the projection formula from linear algebra: x^n is corrected by subtracting its component along each earlier Q_k, the component being $\langle x^n,Q_k\rangle Q_k/h_k$. What remains has no component along any earlier direction, so it is orthogonal to all of them; and because every subtracted vector is already known, nothing is circular.

The same formula, applied repeatedly, is what a computer does when it computes orthogonal polynomials from a weight. Doing it by hand once is instructive.

Example 3 (Legendre polynomials from scratch)**.** *Take $I=(-1,1)$ and $w=1$. Then $Q_0=1$ and $h_0=2$. For $n=1$, $\langle x,1\rangle=0$, so $Q_1=x$ and $h_1=\frac{2}{3}$. For $n=2$, by (10),*

$$Q_2(x)=x^2-\frac{\langle x^2,1\rangle}{2}\cdot 1-\frac{\langle x^2,x\rangle}{2/3}\cdot x=x^2-\frac{2/3}{2}=x^2-\frac{1}{3},$$

since $\langle x^2,x\rangle=0$ and $\langle x^2,1\rangle=\frac{2}{3}$. As a check, $\langle Q_2,1\rangle=\frac{2}{3}-\frac{1}{3}\cdot 2=0$, and $\langle Q_2,x\rangle=0$ because the integrand is odd. Finally

$$h_2=\int_{-1}^{1}\Big(x^2-\frac{1}{3}\Big)^2\mathrm{d}x=\frac{2}{5}-\frac{4}{9}+\frac{2}{9}=\frac{8}{45}.$$

This is the polynomial $x^2 - \frac{1}{3}$ *of* (56).

The construction also shows the following, which is used constantly: Q_n *is orthogonal to every polynomial of degree* $< n$, not only to the other Q_k; and consequently $\langle xQ_n, Q_m\rangle = 0$ *whenever* $m \le n-2$, because then xQ_m has degree $m+1 \le n-1$.

1.4 The three-term recurrence

The most important structural fact about orthogonal polynomials is that each Q_n is a combination of only its two neighbours.

Proposition 4 (Three-term recurrence)**.** *Let* $\{Q_n\}$ *be the monic orthogonal polynomials of a weight* w *and put* $h_n = \langle Q_n, Q_n\rangle$. *Then*

$$xQ_n = Q_{n+1} + b_n Q_n + a_n^2 Q_{n-1}, \qquad n \ge 0, \tag{11}$$

where $Q_{-1} = 0$,

$$b_n = \frac{\langle xQ_n, Q_n\rangle}{h_n}, \qquad a_n^2 = \frac{h_n}{h_{n-1}} \quad (n \ge 1). \tag{12}$$

In particular $a_n^2 > 0$ *and*

$$h_n = a_n^2 h_{n-1}, \qquad h_n = h_0 \prod_{j=1}^{n} a_j^2. \tag{13}$$

Proof. The polynomial xQ_n has degree $n+1$, so it is a linear combination $xQ_n = \sum_{k=0}^{n+1} c_k Q_k$ of $Q_0, \dots, Q_{n+1}$; taking inner products with Q_j and using orthogonality gives $c_j = \langle xQ_n, Q_j\rangle / h_j$. For $j \le n-2$ the polynomial xQ_j has degree at most $n-1$, so $\langle xQ_n, Q_j\rangle = \langle Q_n, xQ_j\rangle = 0$. Thus only c_{n+1}, c_n and c_{n-1} can be nonzero, and $c_{n+1} = 1$ because both sides of the identity are monic of degree $n+1$. This gives (11) with $b_n = c_n$ as in (12), and

$$a_n^2 = c_{n-1} = \frac{\langle xQ_n, Q_{n-1}\rangle}{h_{n-1}} = \frac{\langle Q_n, xQ_{n-1}\rangle}{h_{n-1}} = \frac{\langle Q_n, Q_n\rangle}{h_{n-1}} = \frac{h_n}{h_{n-1}},$$

where the third step uses $xQ_{n-1} = Q_n + (\text{lower degree})$, valid because both polynomials are monic of degree n, and the last step uses orthogonality: the lower-degree part contributes nothing. □

The reason only two neighbours occur is degree counting. Multiplication by x raises every degree by one, so xQ_n of degree $n+1$ can involve Q_{n+1} and lower terms, and orthogonality against Q_m eliminates everything with $m \le n-2$; only $m = n$ and $m = n-1$ survive. The same counting produces the *three*-term form in every family, and it is the source of the tridiagonal matrices of §6.

Two comments. First, the recurrence coefficients are exactly the numbers (12): the recurrence is not an extra structure laid on top of orthogonality, it is equivalent to it. (Conversely, any sequence generated by (11) from $Q_0 = 1$, $Q_1 = x - b_0$ with $a_n^2 > 0$ is orthogonal for some positive weight; this is Favard's theorem, which we do not need but which explains why the associated families of §7 are orthogonal.) Second, in the *standard* normalisation $p_n = k_n Q_n$, where k_n is the leading coefficient, the same identity reads

$$xp_n = \frac{k_n}{k_{n+1}}\, p_{n+1} + b_n p_n + \frac{k_n}{k_{n-1}}\, a_n^2\, p_{n-1}, \tag{14}$$

and this is exactly how one passes from the standard recurrences of §3 to the monic data of Table 2: the coefficient of p_{n+1} is k_n/k_{n+1} by the monicity argument above, the middle coefficient is unchanged, and the last is multiplied by k_n/k_{n-1}.

1.5 Factorials, Pochhammer symbols, gamma and beta

For a non-negative integer n put $n! = 1 \cdot 2 \cdots n$ and $0! = 1$; the binomial coefficient is $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$. The *Pochhammer symbol* (rising factorial) of a number a is

$$(a)_0 = 1, \qquad (a)_n = a(a+1)\cdots(a+n-1) = \frac{\Gamma(a+n)}{\Gamma(a)}, \qquad n \geq 1. \tag{15}$$

For example $(1)_n = n!$, $(2)_n = (n+1)!$, and $(-n)_k = (-1)^k \frac{n!}{(n-k)!}$ for $0 \leq k \leq n$.

The *gamma function* extends the factorial:

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\,\mathrm{d}t \qquad (s > 0), \tag{16}$$

and integration by parts gives $\Gamma(s+1) = s\Gamma(s)$, hence $\Gamma(n+1) = n!$ for integers $n \geq 0$. The two values used below are

$$\Gamma(1) = 1, \qquad \Gamma(\tfrac{1}{2}) = \sqrt{\pi}. \tag{17}$$

The middle expression in (15) is the definition of $\Gamma(a+n)/\Gamma(a)$ for non-integer a; it turns products of consecutive integers, which is what the norms of the classical families are, into single gamma functions.

The *beta function* is

$$B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,\mathrm{d}t = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} \qquad (p, q > 0). \tag{18}$$

Two consequences are used. First, the mass of the Laguerre weight is

$$\int_0^\infty x^\alpha e^{-x}\,\mathrm{d}x = \Gamma(\alpha+1)$$

by (16). Second, substituting $t = (1+x)/2$ gives the mass of the Jacobi weight,

$$\int_{-1}^1 (1-x)^\alpha (1+x)^\beta\,\mathrm{d}x = 2^{\alpha+\beta+1} B(\alpha+1, \beta+1) = \frac{2^{\alpha+\beta+1}\Gamma(\alpha+1)\Gamma(\beta+1)}{\Gamma(\alpha+\beta+2)}, \tag{19}$$

which is the value h_0 in (53) at $n = 0$.

1.6 Facts from calculus

Integration by parts,

$$\int_a^b f'g\,\mathrm{d}x = [fg]_a^b - \int_a^b fg'\,\mathrm{d}x, \tag{20}$$

applied n times to a product whose differentiated factor vanishes at the endpoints together with its derivatives, gives

$$\int_a^b f^{(n)} g\,\mathrm{d}x = (-1)^n \int_a^b f g^{(n)}\,\mathrm{d}x. \tag{21}$$

This is the only integration technique used in the proof of orthogonality from Rodrigues' formula in §2.3.

Differentiation under the integral sign: if $f(t,z)$ is smooth in both variables and the integrals converge uniformly, then

$$\frac{\mathrm{d}}{\mathrm{d}z}\int f(t,z)\,\mathrm{d}t = \int \frac{\partial f}{\partial z}(t,z)\,\mathrm{d}t. \tag{22}$$

We use it to differentiate $\frac{1}{z-t}$ with respect to z, which is legitimate for z away from the integration interval.

The last fact underlies the Stieltjes transform. Let ϕ be continuous on an interval containing x. Then

$$\lim_{y\downarrow 0}\int \frac{y}{(x-t)^2+y^2}\,\phi(t)\,\mathrm{d}t = \pi\phi(x). \tag{23}$$

The kernel $y/((x-t)^2+y^2)$ is positive, its total integral is π (substitute $t = x+yu$ and use $\int \mathrm{d}u/(1+u^2)=\pi$), and it concentrates near $t=x$ as $y\downarrow 0$: for $|t-x|\geq\delta$ it is at most y/δ^2, which tends to zero, while on $|t-x|<\delta$ the function $\phi(t)$ differs from $\phi(x)$ by an arbitrarily small amount once δ is small. This is the first-course computation behind the statement that a narrow kernel picks out the value of a function at the centre; applied to a density $\nu'(t)$ instead of a function, it gives $\pi\nu'(x)$, and this is how the identity is used in §6.

1.7 Complex numbers and branches

A complex number is $z = x+\mathrm{i}y$ with real x and y; $\operatorname{Re} z = x$, $\operatorname{Im} z = y$, $\overline{z} = x-\mathrm{i}y$ and $|z|^2 = z\overline{z} = x^2+y^2$. Also

$$\frac{1}{z} = \frac{\overline{z}}{|z|^2}, \qquad \operatorname{Im}\frac{1}{z} = -\frac{\operatorname{Im} z}{|z|^2}. \tag{24}$$

The upper half-plane is the set of z with $\operatorname{Im} z>0$; the integrals of §6 are taken there, and the limit $\operatorname{Im} z\downarrow 0$ is written $z = x+\mathrm{i}0$.

The logarithm needs a branch. On the principal branch $\ln z = \ln|z| + \mathrm{i}\arg z$ with $-\pi<\arg z\leq\pi$. A point just above the negative real axis therefore has argument close to π, so

$$\ln(-x+\mathrm{i}0) = \ln x + \mathrm{i}\pi \qquad (x>0), \tag{25}$$

while a point just below it has argument close to $-\pi$ and $\ln(-x-\mathrm{i}0) = \ln x - \mathrm{i}\pi$. This is the only branch fact used: it is what turns $\ln\frac{z+1}{z-1}$ into $\ln\frac{1+x}{1-x}-\mathrm{i}\pi$ for $-1<x<1$ in §7. The sign is easy to get wrong, so here is the check. Take $z = x+\mathrm{i}y$ with $y>0$ and $-1<x<1$. Then $z+1 = (x+1)+\mathrm{i}y$ lies in the right half-plane, so its logarithm has no $\mathrm{i}\pi$; but $z-1 = (x-1)+\mathrm{i}y$ has negative real part and positive imaginary part, so it lies just above the negative real axis and its logarithm carries $+\mathrm{i}\pi$. Therefore

$$\ln\frac{z+1}{z-1} \longrightarrow \ln\frac{1+x}{1-x} - \mathrm{i}\pi \qquad (y\downarrow 0),$$

the minus sign coming from the denominator, not from a choice of branch below the axis.

Square roots of positive numbers are taken positive throughout.

1.8 Matrices, determinants and eigenvalues

A real *symmetric* matrix M ($M^{T} = M$) has real eigenvalues and an orthonormal basis of eigenvectors. Its *characteristic polynomial* is $\det(xI - M)$, a monic polynomial of degree equal to the size of the matrix whose roots are the eigenvalues.

For a tridiagonal matrix

$$J_n = \begin{pmatrix} b_0 & a_1 & & \\ a_1 & b_1 & a_2 & \\ & a_2 & b_2 & \ddots \\ & & \ddots & \ddots \end{pmatrix}, \qquad a_j > 0,$$

expand $\det(xI - J_n)$ along the last row. That row has only two nonzero entries: the diagonal entry $x - b_{n-1}$ and the subdiagonal entry $-a_{n-1}$. The diagonal term is $(x - b_{n-1})Q_{n-1}(x)$, since deleting the last row and column leaves the same tridiagonal determinant with $n - 1$ rows. For the other term, delete the last row and the next-to-last column; the remaining determinant has a last column with the single nonzero entry $-a_{n-1}$, and expanding along that column gives $(-a_{n-1})Q_{n-2}(x)$ times the checkerboard sign. That sign is -1 for the row expansion, so the two minus signs cancel and the contribution is $-a_{n-1}^2 Q_{n-2}(x)$. This gives

$$Q_n(x) = (x - b_{n-1})Q_{n-1}(x) - a_{n-1}^2 Q_{n-2}(x), \qquad Q_0 = 1, \quad Q_1 = x - b_0, \tag{26}$$

which is (11) read backwards. For $n = 2$ and $n = 3$ this is the familiar computation

$$Q_2 = (x - b_1)(x - b_0) - a_1^2, \qquad Q_3 = (x - b_2)Q_2 - a_2^2(x - b_0),$$

and it shows that *the monic orthogonal polynomials are the characteristic polynomials of the finite sections of the tridiagonal matrix with diagonal b_n and off-diagonal a_{n+1}*. This is the bridge between the analysis of §3 and the spectral picture of §6.

The infinite matrix J with $J_{nn} = b_n$ and $J_{n,n+1} = J_{n+1,n} = a_{n+1}$ is the *Jacobi matrix*. An infinite matrix is not a matrix in the first-year sense, and nothing below needs one: every statement about J is a statement about its finite sections J_n, made for all n and then passed to a limit. Concretely, fix k and take $n > k$. The matrix J_n maps the span of $e_0, \dots, e_k$ into itself, so the $(1,1)$ entry of J_n^k does not depend on n; call it m_k. Thus

$$m_k = (e_0, J_n^k e_0) \qquad (n > k), \qquad k = 0, 1, 2, \dots, \tag{27}$$

where $e_0 = (1, 0, \dots, 0)$ and J_n^k is the k-fold product of the finite matrix with itself. In particular $m_0 = 1$. The numbers m_k are the moments of the weight: §6 shows from the recurrence that $m_k = \int t^k w(t)\,\mathrm{d}t/h_0$. A *limit mass distribution* ν is the object whose moments are the m_k; §6 builds it as the limit of the finite spectral data of J_n. No infinite matrix is inverted or exponentiated anywhere in this note.

1.9 Two special functions

Two non-elementary functions appear in the closed forms for the Stieltjes transforms.

The *exponential integral* is, for real $x > 0$,

$$\mathrm{Ei}(x) = \gamma + \ln x + \sum_{k=1}^{\infty} \frac{x^k}{k \cdot k!} = -\int_{-x}^{\infty} \frac{e^{-t}}{t}\,\mathrm{d}t \quad \text{(principal value)}, \tag{28}$$

where $\gamma = 0.5772\dots$ is Euler's constant and the last integral is a principal value. The analytic continuation to the upper half-plane is given by the same series with ln on the principal branch.

The *complementary error function* is

$$\operatorname{erfc}(z) = 1 - \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-t^2}\,\mathrm{d}t, \qquad \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\,\mathrm{d}t. \tag{29}$$

It is entire, with $\operatorname{erfc}(0) = 1$ and $\operatorname{erfc}(z) \to 0$ as $z \to +\infty$ along the real axis.

Two integral representations are used, each for $\operatorname{Im} z > 0$ and each checked numerically in §6:

$$\int_0^\infty \frac{e^{-t}}{z-t}\,\mathrm{d}t = e^{-z}\bigl(\operatorname{Ei}(z) - \mathrm{i}\pi\bigr), \qquad \int_{-\infty}^\infty \frac{e^{-t^2}}{z-t}\,\mathrm{d}t = -\mathrm{i}\pi e^{-z^2}\operatorname{erfc}(-\mathrm{i}z). \tag{30}$$

For the first, substitute $t = z + s$; the integral becomes $-e^{-z}\int_{-z}^\infty e^{-s}/s\,\mathrm{d}s$, and the relation between the exponential integral and this integral gives the stated form. The second is the standard integral representation of erfc, obtained by completing the square; a reader who wants to check it without that machinery can take it as the definition of the analytic continuation and verify it against numerical quadrature, which is what is done in §6.

2 What "classical" means

2.1 The Bochner problem

Every family in this note is a sequence of polynomials $p_0, p_1, p_2, \dots$ with $\deg p_n = n$. Orthogonal polynomials are usually defined through a weight, but the classical families can be characterised without one, and the characterisation is the reason they are called classical.

Definition 5. *A sequence* $\{p_n\}$, $\deg p_n = n$, *is* classical *if there are polynomials* σ *and* τ, *independent of* n, *with* $\sigma \not\equiv 0$, *and constants* λ_n, *such that*

$$\sigma(x)p_n''(x) + \tau(x)p_n'(x) + \lambda_n p_n(x) = 0 \qquad (n = 0, 1, 2, \dots). \tag{31}$$

The point of the definition is that σ and τ do not depend on n: the equation has the same shape for every member of the family and only the number λ_n changes. Equation (31) is a *second-order linear differential equation*: the unknown p_n and its first two derivatives appear linearly, with polynomial coefficients. For fixed n its solutions form a two-dimensional space; the classical polynomials are the polynomial solutions, and the polynomial solution of degree n is unique up to multiplication by a constant.

Geometrically, $\sigma p_n''$ controls how sharply p_n bends, $\tau p_n'$ controls its growth or decay, and $\lambda_n p_n$ rescales it. Asking the same two coefficient functions σ, τ to do this for every degree n is a strong demand: it forces σ to be at most quadratic and τ at most linear, as the theorem below shows. That is the content of "classical".

Example 6. *For Hermite,* $\sigma = 1$ *and* $\tau = -2x$, *so* (31) *reads* $p'' - 2xp' + \lambda_n p = 0$ *with* $\lambda_n = 2n$. *For* $n = 1$ *and* $p = 2x$: $p'' = 0$, $p' = 2$, *and* $0 - 2x \cdot 2 + 2 \cdot 2x = 0$. *For* $n = 2$ *and* $p = 4x^2 - 2$: $p'' = 8$, $p' = 8x$, *and* $8 - 16x^2 + 4(4x^2 - 2) = 0$. *The same* σ *and* τ *work for every* n; *only* λ_n *changes.*

Bochner [2] asked which families have this property, and the answer is short. The theorem is usually stated for families that are orthogonal with respect to a positive weight, and we state it that way: the weight is what rules out a fifth case.

Theorem 7 (Bochner, 1929)**.** *Let $\{p_n\}$ be orthogonal polynomials with respect to a positive weight w on an interval of the real line, and suppose that they satisfy (31) with $\deg p_n = n$ and σ, τ polynomials independent of n. Then*

$$\deg \sigma \le 2, \qquad \deg \tau \le 1, \qquad \lambda_n = -\frac{n(n-1)}{2}\sigma'' - n\tau', \tag{32}$$

and, after an affine change of the variable and multiplication by a constant, (σ, τ, w) is one of the four rows of Table 1.

Proof. The proof is only degree bookkeeping: look at the highest power that cannot be cancelled. Write $p_n(x) = x^n + c_1 x^{n-1} + \cdots$ and let $d = \deg \sigma$ and $e = \deg \tau$. Fix $n \ge 2$. The three terms of (31) have degrees at most $d+n-2$, $e+n-1$ and n, and the coefficients of the highest powers are

$$\sigma p_n'' \sim d\, n(n-1)\sigma_d\, x^{d+n-2}, \qquad \tau p_n' \sim e\, n\, \tau_e\, x^{e+n-1}, \qquad \lambda_n p_n \sim \lambda_n x^n.$$

The term $\lambda_n p_n$ has degree at most n (at most, because λ_n may vanish), so every term of degree $> n$ has to be cancelled by another term of the same degree.

Step 1: $e \le 1$. Suppose $e \ge 2$. Then $\tau p_n'$ has degree $e+n-1 \ge n+1 > n$, which the λ term cannot touch. If $e \ge d$ then $e+n-1 > d+n-2$, so $\tau p_n'$ is the unique term of degree $e+n-1$, with coefficient $en\tau_e \ne 0$, and the sum cannot vanish. If $e \le d-1$ the highest degree is $D = d+n-2 \ge e+n-1 > n$. When $d \ge e+2$ the term $\sigma p_n''$ alone has degree D, with coefficient $dn(n-1)\sigma_d \ne 0$. When $d = e+1$ the two terms share the degree $D = e+n-1 = d+n-2$, and their coefficients add to

$$dn(n-1)\sigma_d + en\tau_e = n\big[(n-1)(e+1)\sigma_{e+1} + e\tau_e\big],$$

a nonzero linear function of n, so the sum cannot vanish for every $n \ge 2$. Hence $e \ge 2$ is impossible and $e \le 1$.

Step 2: $d \le 2$. With $e \le 1$, the terms $\tau p_n'$ and $\lambda_n p_n$ have degree at most n. If $d \ge 3$, then $\sigma p_n''$ has degree $D = d+n-2 \ge n+1 > n$, strictly larger than the other two degrees, and its coefficient $dn(n-1)\sigma_d$ is nonzero. The equation cannot hold, so $d \le 2$.

Step 3: the eigenvalue. With $d \le 2$ and $e \le 1$, the coefficient of x^n, for $n \ge 2$, is $\sigma_2 n(n-1) + \tau_1 n + \lambda_n$ (with $\sigma_2 = 0$ if $\deg\sigma < 2$ and $\tau_1 = 0$ if $\deg\tau < 1$). It must vanish, so

$$\lambda_n = -\frac{n(n-1)}{2}\,\sigma'' - n\tau',$$

which is (32). For $n = 1$ the coefficient of x gives $\tau_1 + \lambda_1 = 0$, which is the same formula, and $\lambda_0 = 0$ because p_0 is a nonzero constant.

Step 4: the four rows. A polynomial of degree at most two has at most two roots, and an affine change of the variable brings σ to one of the forms

$$1 - x^2 \text{ (two distinct real roots)}, \quad x^2 \text{ (a double root)}, \quad x \text{ (degree one, one root)}, \quad 1 \text{ (constant)}.$$

A real quadratic with no real root is the fifth algebraic possibility. It is not on this list because it has no root to move to the origin; it normalises to $1 + x^2$ and is treated separately below.

In each case the Pearson equation $(\sigma w)' = \tau w$ of §2.2 determines w, and the requirement that w be positive, integrable and have all moments restricts τ to the parameter range of the corresponding

row of Table 1. For $\sigma = 1 - x^2$ and $\sigma = x$ these are the computations at the end of §2.2, which give Jacobi and Laguerre. For $\sigma = 1$ the weight is $w = e^{\tau_1 x^2/2 + \tau_0 x}$, integrable on $\mathbb{R}$ only when $\tau_1 < 0$; an affine change of variable then brings τ to $-2x$, which is Hermite. For $\sigma = x^2$ the moments fail to exist (Bessel subsection), so no positive weight on an interval of the line arises; this is the Bessel row, orthogonal only on the unit circle.

The case of a quadratic σ with no real root is the fifth algebraic possibility, and it produces no orthogonal family. Such a σ normalises to $1 + x^2$. The Pearson weight is

$$w(x) \propto (1 + x^2)^{(\tau_1 - 2)/2} e^{\tau_0 \arctan x},$$

because $w'/w = (\tau_1 - 2)x/(1 + x^2) + \tau_0/(1 + x^2)$. As $|x| \to \infty$ we have $\arctan x \to \pm\pi/2$, so $w(x) = c_\pm\, |x|^{\tau_1 - 2}\,(1 + o(1))$ with $c_\pm = e^{\pm\tau_0 \pi/2} > 0$. Consequently the moment $\int |t|^k\, w(t)\, \mathrm{d}t$ is finite only for $k < 1 - \tau_1$, whatever τ_1 is; some moment always diverges. Orthogonality with respect to a positive weight requires every moment to be finite, the first ones being those that define $p_1 = x - c$ and p_2, so no choice of τ gives an orthogonal family. (Equivalently, the boundary terms of §2.5 behave like $|x|^{\tau_1 + n + m - 2}$ and do not vanish for large n, m.) This is why the table has four rows and not five: it is integrability of the weight, not the degree count, that removes the fifth case. □

The theorem is the reason there are no other classical families: the degree count leaves only the forms above, and positivity and integrability of the weight remove the remaining possibilities.

2.2 The Pearson equation

The weight is recovered from σ and τ by a first-order equation. Write the differential equation (31) in Sturm–Liouville form: it is equivalent to

$$\big(\sigma w\, p_n'\big)' + \lambda_n\, w\, p_n = 0, \tag{33}$$

provided

$$(\sigma w)' = \tau w. \tag{34}$$

Indeed, expanding (33) and dividing by w gives $\sigma p_n'' + (\sigma w'/w + \sigma')p_n' + \lambda_n p_n = 0$, which is (31) exactly when $\tau = \sigma' + \sigma w'/w$.

Equation (34) is the *Pearson equation*. Expanded, it reads $\sigma w' + \sigma' w = \tau w$, that is

$$\frac{w'}{w} = \frac{\tau - \sigma'}{\sigma}, \tag{35}$$

a *separable* first-order equation: the left side is a derivative, $w'/w = (\ln w)'$, and the right side depends on x only, so $\ln w = \int (\tau - \sigma')/\sigma \, \mathrm{d}x$. In words: the logarithmic derivative of the weight is a rational function whose denominator is σ. That is exactly the structure that makes the combination $\sigma w p_n'$ telescope under integration by parts, and it is why the weight can be read off from the differential equation. In each of the three cases the integral is elementary:

- $\sigma = 1 - x^2$, $\tau = \beta - \alpha - (\alpha + \beta + 2)x$:

$$\frac{w'}{w} = \frac{\beta - \alpha - (\alpha + \beta)x}{1 - x^2} = -\frac{\alpha}{1 - x} + \frac{\beta}{1 + x}, \qquad w = (1 - x)^\alpha (1 + x)^\beta.$$

- $\sigma = x$, $\tau = \alpha + 1 - x$:
$$\frac{w'}{w} = \frac{\alpha - x}{x} = \frac{\alpha}{x} - 1, \qquad w = x^{\alpha} e^{-x}.$$
- $\sigma = 1$, $\tau = -2x$:
$$\frac{w'}{w} = -2x, \qquad w = e^{-x^2}.$$

The ranges $\alpha, \beta > -1$ and $\alpha > -1$ are exactly the ranges in which these weights are integrable at the endpoints.

2.3 The Rodrigues formula

The third identity in (2) is a consequence of the Pearson equation. Define, for $n \geq 0$,

$$u_n(x) = \frac{\mathrm{d}^n}{\mathrm{d}x^n}\big(\sigma(x)^n w(x)\big), \qquad p_n(x) = \frac{u_n(x)}{w(x)} = \frac{1}{w}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\big(\sigma^n w\big). \tag{36}$$

We prove that p_n is a polynomial of degree n, that it satisfies (31) with the eigenvalue (32), and that the family is orthogonal with weight w. The proof has six steps and uses nothing beyond the product rule and the Pearson equation. The plan is to differentiate the first-order equation for $\sigma^n w$ exactly n times; because σ has degree at most two and τ at most one, the Leibniz rule stops after three terms, and what remains is the second-order equation for p_n.

Step 1: p_n is a polynomial of degree n. For the three families $\sigma^n w = w\sigma^n$, and the quotient (36) is a polynomial for the following reason. By the Pearson equation $w'/w = (\tau - \sigma')/\sigma$, so each derivative $w^{(k)} = w\rho_k$ has ρ_k a rational function whose denominator divides σ^k: the case $k = 0$ is clear, and differentiating $w^{(k)} = w\rho_k$ shows that the denominator of ρ_{k+1} divides σ^{k+1}. On the other hand $\dfrac{\mathrm{d}^{n-k}}{\mathrm{d}x^{n-k}}\sigma^n$ is divisible by σ^k (write $\sigma^n = \sigma^k \sigma^{n-k}$ and use the product rule). In the Leibniz expansion

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}(w\sigma^n) = \sum_{k=0}^{n} \binom{n}{k} w^{(k)} \frac{\mathrm{d}^{n-k}}{\mathrm{d}x^{n-k}}\sigma^n$$

the kth term is therefore w times a polynomial, so the sum is wR_n with R_n a polynomial. For the three families R_n has degree exactly n; this is what the explicit formulas of §3 show, and the leading coefficients are those of Table 2.

Step 2: a first-order equation for $f = \sigma^n w$. Since $\sigma^{n+1} w = \sigma f$ and $\sigma w' = \tau w - \sigma' w$ by the Pearson equation, the product rule gives

$$(\sigma f)' = (n+1)\sigma^n \sigma' w + \sigma^{n+1} w' = \sigma^n\big((n+1)\sigma' + \tau - \sigma'\big)w = (\tau + n\sigma')f. \tag{37}$$

Subtracting $\sigma' f$ from both sides of $(\sigma f)' = \sigma' f + \sigma f'$ gives

$$\sigma f' = \big(\tau + (n-1)\sigma'\big) f. \tag{38}$$

Write $R = \tau + (n-1)\sigma'$; this has degree at most one, because $\deg \tau \leq 1$ and $\deg \sigma \leq 2$.

Step 3: differentiate (38) n times. For σ of degree at most two the Leibniz rule keeps the terms with $k = 0, 1, 2$, and for R of degree at most one the terms with $k = 0, 1$:

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}(\sigma f') = \sigma f^{(n+1)} + n\sigma' f^{(n)} + \frac{n(n-1)}{2}\sigma'' f^{(n-1)}, \qquad \frac{\mathrm{d}^n}{\mathrm{d}x^n}(Rf) = Rf^{(n)} + nR' f^{(n-1)}.$$

Equating the two right-hand sides and putting $u = f^{(n)}$ gives

$$\sigma u' + (\sigma' - \tau)u + Cf^{(n-1)} = 0, \qquad C = \frac{n(n-1)}{2}\sigma'' - nR', \tag{39}$$

where $n\sigma' - R = \sigma' - \tau$ was used.

Step 4: differentiate (39) once more. The term $f^{(n-1)}$ differentiates into u, and substituting $R' = \tau' + (n-1)\sigma''$ gives

$$C = \frac{n(n-1)}{2}\sigma'' - n\tau' - n(n-1)\sigma'' = -\frac{n(n-1)}{2}\sigma'' - n\tau' = \lambda_n.$$

Therefore

$$\sigma u'' + (2\sigma' - \tau)u' + (\sigma'' - \tau' + \lambda_n)u = 0. \tag{40}$$

Step 5: divide by w. Since $u = wp_n$, differentiating $p_n = u/w$ twice gives

$$p_n' = \frac{u'}{w} - \frac{w'}{w^2}u, \qquad p_n'' = \frac{u''}{w} - 2\frac{w'}{w^2}u' - \frac{w''}{w^2}u + 2\frac{(w')^2}{w^3}u.$$

Inserting these and $w'/w = (\tau - \sigma')/\sigma$ into the left side of

$$\sigma w p_n'' + \tau w p_n' + \lambda_n w p_n = \sigma u'' + (2\sigma' - \tau)u' + (\sigma'' - \tau' + \lambda_n)u \tag{41}$$

gives $\sigma u'' + (2\sigma' - \tau)u' + Bu$, where

$$B = -\sigma\frac{w''}{w} + 2\sigma\Big(\frac{w'}{w}\Big)^2 - \tau\frac{w'}{w} + \lambda_n.$$

Writing $S = w'/w = (\tau - \sigma')/\sigma$, one has $\sigma S' = \tau' - \sigma'' - \sigma' S$ and $\sigma S^2 - \tau S = -\sigma' S$, so $B = -\sigma S' + \sigma S^2 - \tau S + \lambda_n = \sigma'' - \tau' + \lambda_n$. This proves (41); together with (40) it gives the differential equation (31) for p_n with the eigenvalue (32).

Step 6: orthogonality. For $m < n$, integration by parts n times using (21) gives

$$\langle p_n, p_m\rangle = \int \frac{1}{w}\big(\sigma^n w\big)^{(n)} p_m\, w\,\mathrm{d}x = \int \big(\sigma^n w\big)^{(n)} p_m\,\mathrm{d}x = (-1)^n \int \sigma^n w p_m^{(n)}\,\mathrm{d}x = 0,$$

because $p_m^{(n)} = 0$. The boundary terms are $\big[\frac{\mathrm{d}^k}{\mathrm{d}x^k}(\sigma^n w)\, p_m^{(n-1-k)}\big]$ with $0 \le k \le n-1$, and they vanish: at each end of the interval either σ vanishes (Jacobi at $x = \pm 1$, Laguerre at $x = 0$) or the weight does (Hermite at infinity), and every term of the kth derivative still contains a positive power of the vanishing factor. Explicitly, for Jacobi every term of the kth derivative is $(1-x)^{n+\alpha-a}(1+x)^{n+\beta-b}$ with $a + b = k \le n-1$, whose exponents are at least $\alpha + 1$ and $\beta + 1$; for Laguerre every term is

$x^{n+\alpha-a}e^{-x}$ with $a \le n-1$, whose exponent is at least $\alpha+1$; for Hermite every term is a polynomial times e^{-x^2}.

The standard normalisations are the choices of the constant factor for which the formulas read conventionally; in the three cases the Rodrigues formula is

$$P_n^{(\alpha,\beta)}(x) = \frac{(-1)^n}{2^n n!}(1-x)^{-\alpha}(1+x)^{-\beta}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\Big[(1-x)^{n+\alpha}(1+x)^{n+\beta}\Big], \tag{42}$$

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha}e^x}{n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\big(x^{n+\alpha}e^{-x}\big), \tag{43}$$

$$H_n(x) = (-1)^n e^{x^2}\frac{\mathrm{d}^n}{\mathrm{d}x^n}e^{-x^2}. \tag{44}$$

2.4 The fourth family: Bessel

The case $\sigma = x^2$ is the one that behaves differently. The Pearson equation gives

$$(x^2 w)' = (ax+b)w \quad\Longrightarrow\quad w(x) \propto x^{a-2}e^{-b/x},$$

which is positive on $(0,\infty)$, but is not a usable weight there: as $x\to\infty$ it behaves like x^{a-2}, so $\int_0^\infty x^k w\,\mathrm{d}x$ diverges once $k \ge 1-a$, and the moments that orthogonality needs do not exist. On an interval containing negative values of x the factor x^{a-2} is not real-valued when a is not an integer, and it changes sign for integer a, so positivity fails there too. Thus there is no positive weight on an interval of the real line with respect to which these polynomials are orthogonal; the corresponding polynomials are the *Bessel polynomials* $y_n(x;a,b)$, defined as the polynomial solutions of

$$x^2 y_n'' + (ax+b)y_n' - n(n+a-1)y_n = 0, \qquad y_n(0) = 1. \tag{45}$$

They are polynomials, not infinite series, for the following reason. Write $y_n = \sum_{k\ge0} c_k x^k$ with $c_0 = 1$ and substitute. The coefficient of x^k is

$$\big[k(k-1) + ak - n(n+a-1)\big]c_k + b(k+1)c_{k+1} = 0,$$

and the bracket factors as $(k-n)(k+n+a-1)$. Hence

$$c_{k+1} = -\frac{(k-n)(k+n+a-1)}{b(k+1)}\,c_k,$$

so $c_{n+1} = 0$: the series terminates and y_n has degree n. For example $y_0 = 1$, $y_1 = 1 + \frac{a}{b}x$, and $y_2 = 1 + \frac{2(a+1)}{b}x + \frac{(a+1)(a+2)}{b^2}x^2$. They are orthogonal on the unit circle, with respect to a weight built from gamma functions, but not on the real line with respect to a positive weight. For this reason the three positive-definite families — Jacobi, Laguerre, Hermite — are what one means by "the" classical orthogonal polynomials, and Bessel is the fourth member of Bochner's classification that one sets aside. The rest of the note is about the three.

2.5 Why the differential equation gives orthogonality

It is worth recording that orthogonality follows from (33) alone, without knowing the weight explicitly. Suppose $w > 0$ and $\sigma w \to 0$ at both ends of the interval. Multiply (33) for p_n by p_m, integrate, and integrate by parts:

$$0 = \int (\sigma w p_n')' p_m\,\mathrm{d}x + \lambda_n \int w p_n p_m\,\mathrm{d}x = \underbrace{[\sigma w p_n' p_m]}_{\text{boundary}} - \int \sigma w p_n' p_m'\,\mathrm{d}x + \lambda_n \int w p_n p_m\,\mathrm{d}x,$$

and the boundary term vanishes by the hypothesis on σw. The remaining integral is symmetric in m and n; subtracting the same identity with m and n interchanged therefore gives

$$(\lambda_n - \lambda_m) \int w p_n p_m \, \mathrm{d}x = 0.$$

When $\lambda_n \neq \lambda_m$ the integral vanishes. By (32) the eigenvalues are easy to read off. For Jacobi, $\sigma'' = -2$ and $\tau' = -(\alpha + \beta + 2)$, so

$$\lambda_n = -\frac{n(n-1)}{2}(-2) - n\big(-(\alpha + \beta + 2)\big) = n(n + \alpha + \beta + 1).$$

For Laguerre, $\sigma'' = 0$ and $\tau' = -1$, so $\lambda_n = n$; for Hermite, $\sigma'' = 0$ and $\tau' = -2$, so $\lambda_n = 2n$. The first is strictly increasing in n because $\alpha + \beta + 2 > 0$, while $\lambda_n = n$ and $\lambda_n = 2n$ are strictly increasing as well; so the eigenvalues of a family are pairwise distinct and the family is orthogonal.

This argument is the model for everything that follows: the differential equation is powerful precisely because it is independent of n except for the eigenvalue. It also explains the failure of the case $\sigma = 1 + x^2$ discussed after Bochner's theorem: there $\sigma w p_n' p_m$ behaves like $|x|^{\tau_1 + n + m - 2}$ at infinity, which does not tend to zero once $n + m > 2 - \tau_1$, and consistently the Pearson weight has divergent moments.

3 The three families

In this section we treat the three families one after another. For each we record the weight, the standard normalisation, the differential equation, the Rodrigues formula, the standard three-term recurrence and the leading coefficient, and we then pass to the monic normalisation. The passage is mechanical; it is carried out in full for Jacobi and briefly for the other two, so that every entry of Table 2 can be checked by the reader.

Lemma 8 (Standard recurrence to monic recurrence)**.** *Suppose the standard normalisation $p_n = k_n Q_n$ satisfies*

$$x p_n = A_n p_{n+1} + B_n p_n + C_n p_{n-1}$$

with A_n, C_n of the same sign. Then $x Q_n = Q_{n+1} + b_n Q_n + a_n^2 Q_{n-1}$ with

$$b_n = B_n, \qquad a_n^2 = C_n \frac{k_{n-1}}{k_n}, \tag{46}$$

and necessarily $A_n = k_n / k_{n+1}$.

Proof. Substituting $p_n = k_n Q_n$ gives $k_n x Q_n = A_n k_{n+1} Q_{n+1} + B_n k_n Q_n + C_n k_{n-1} Q_{n-1}$. Dividing by k_n,

$$x Q_n = \frac{A_n k_{n+1}}{k_n} Q_{n+1} + B_n Q_n + \frac{C_n k_{n-1}}{k_n} Q_{n-1}.$$

Both $x Q_n$ and Q_{n+1} are monic of degree $n+1$, so the coefficient in front of Q_{n+1} must be 1, that is $A_n k_{n+1} / k_n = 1$; this identifies A_n and leaves the other two coefficients as stated. Since $a_n^2 > 0$, the sign condition on A_n and C_n together with the sign of k_{n-1}/k_n produces a positive a_n^2; for Laguerre, where k_n alternates sign, both A_n and C_n are negative. □

3.1 Jacobi

The Jacobi polynomials are orthogonal on $(-1,1)$ with weight $w=(1-x)^{\alpha}(1+x)^{\beta}$, $\alpha,\beta>-1$. The standard normalisation $P_n^{(\alpha,\beta)}$ has the differential equation

$$(1-x^2)y''+\big(\beta-\alpha-(\alpha+\beta+2)x\big)y'+n(n+\alpha+\beta+1)y=0, \tag{47}$$

the Rodrigues formula (42), and the leading coefficient

$$k_n=\frac{(n+\alpha+\beta+1)_n}{2^n n!}, \tag{48}$$

so that $P_n^{(\alpha,\beta)}=k_n x^n+\dots$ and $Q_n=P_n/k_n$ is monic. The standard three-term recurrence is

$$\begin{aligned}&2(n+1)(n+\alpha+\beta+1)(2n+\alpha+\beta)\,P_{n+1}^{(\alpha,\beta)}(x)\\&\quad=(2n+\alpha+\beta+1)\Big[(2n+\alpha+\beta)(2n+\alpha+\beta+2)x+\alpha^2-\beta^2\Big]P_n^{(\alpha,\beta)}(x)\\&\qquad-2(n+\alpha)(n+\beta)(2n+\alpha+\beta+2)\,P_{n-1}^{(\alpha,\beta)}(x).\end{aligned} \tag{49}$$

Divide (49) by the coefficient of $P_n^{(\alpha,\beta)}$ on its right-hand side and compare with $xp_n=A_np_{n+1}+B_np_n+C_np_{n-1}$. The result is

$$\begin{aligned}A_n&=\frac{2(n+1)(n+\alpha+\beta+1)}{(2n+\alpha+\beta+1)(2n+\alpha+\beta+2)},\\B_n&=\frac{\beta^2-\alpha^2}{(2n+\alpha+\beta)(2n+\alpha+\beta+2)},\qquad C_n=\frac{2(n+\alpha)(n+\beta)}{(2n+\alpha+\beta+1)(2n+\alpha+\beta)}.\end{aligned} \tag{50}$$

The two ratios of leading coefficients needed in (46) are

$$\frac{k_n}{k_{n+1}}=\frac{2(n+1)(n+\alpha+\beta+1)}{(2n+\alpha+\beta+1)(2n+\alpha+\beta+2)},\qquad \frac{k_{n-1}}{k_n}=\frac{2n(n+\alpha+\beta)}{(2n+\alpha+\beta-1)(2n+\alpha+\beta)},$$

and the first is exactly A_n, as the lemma requires. Substituting the second in (46) gives the monic recurrence (3) with

$$b_n=\frac{\beta^2-\alpha^2}{(2n+\alpha+\beta)(2n+\alpha+\beta+2)}, \tag{51}$$

$$\begin{aligned}a_n^2&=\frac{2(n+\alpha)(n+\beta)}{(2n+\alpha+\beta+1)(2n+\alpha+\beta)}\cdot\frac{2n(n+\alpha+\beta)}{(2n+\alpha+\beta-1)(2n+\alpha+\beta)}\\&=\frac{4n(n+\alpha)(n+\beta)(n+\alpha+\beta)}{(2n+\alpha+\beta)^2(2n+\alpha+\beta+1)(2n+\alpha+\beta-1)}.\end{aligned} \tag{52}$$

At $n=0$ the denominator of (51) vanishes when $\alpha+\beta=0$; the formula is then read as the limit $b_0=(\beta-\alpha)/(\alpha+\beta+2)$, which is the value obtained from the recurrence itself. The norm is

$$h_n^{(\alpha,\beta)}=\frac{2^{\alpha+\beta+1}}{2n+\alpha+\beta+1}\,\frac{\Gamma(n+\alpha+1)\Gamma(n+\beta+1)}{\Gamma(n+\alpha+\beta+1)\,n!}, \tag{53}$$

in the standard normalisation; the monic norm is h_n/k_n^2. This is not independent data. The monic norm is forced by (13), $h_n^{\mathrm{mon}}=h_0\prod_{j=1}^{n}a_j^2$, with h_0 the mass (19); the product telescopes, and the displayed gamma expression is its evaluation.

The first polynomials in the monic normalisation are

$$Q_0 = 1, \qquad Q_1 = x + \frac{\alpha - \beta}{\alpha + \beta + 2}, \qquad Q_2 = x^2 - (b_0 + b_1)x + b_0 b_1 - a_1^2. \tag{54}$$

The first of these is worth checking by hand: the standard polynomial is $P_1^{(\alpha,\beta)} = \tfrac{1}{2}\big[(\alpha - \beta) + (\alpha + \beta + 2)x\big]$ and $k_1 = (\alpha + \beta + 2)/2$, so P_1/k_1 is indeed $x + (\alpha - \beta)/(\alpha + \beta + 2)$.

The family is symmetric under $(\alpha, \beta) \leftrightarrow (\beta, \alpha)$ and $x \to -x$:

$$P_n^{(\alpha,\beta)}(-x) = (-1)^n P_n^{(\beta,\alpha)}(x). \tag{55}$$

The Legendre case. At $\alpha = \beta = 0$ the weight is 1, $b_n = 0$, $a_n^2 = n^2/(4n^2 - 1)$, and

$$Q_0 = 1, \quad Q_1 = x, \quad Q_2 = x^2 - \frac{1}{3}, \quad Q_3 = x^3 - \frac{3}{5}x, \qquad h_n^{\mathrm{mon}} = \frac{2}{2n+1}\frac{1}{k_n^2}. \tag{56}$$

The polynomials here are exactly the ones computed by hand in Example 3, and the norm identity (13) can be checked on them: $h_1 = a_1^2 h_0 = \frac{1}{3} \cdot 2 = \frac{2}{3}$, which is $\int_{-1}^{1} x^2 \, \mathrm{d}x$, and $h_2 = a_2^2 h_1 = \frac{4}{15} \cdot \frac{2}{3} = \frac{8}{45}$, which is the value found in that example. This is the family for which the associated case has the longest history: the "associated Legendre polynomials" of §7 are the shifted Legendre family, and their weight $6/(\pi^2 + \ln^2 \frac{1+x}{1-x})$ is one of the standard examples in the theory of orthogonal polynomials.

3.2 Laguerre

The Laguerre polynomials are orthogonal on $(0, \infty)$ with weight $w = x^\alpha e^{-x}$, $\alpha > -1$. The standard normalisation $L_n^{(\alpha)}$ has

$$xy'' + (\alpha + 1 - x)y' + ny = 0, \tag{57}$$

the Rodrigues formula (43), the leading coefficient $k_n = (-1)^n/n!$, and the recurrence

$$(n+1)L_{n+1}^{(\alpha)} = (2n + \alpha + 1 - x)L_n^{(\alpha)} - (n + \alpha)L_{n-1}^{(\alpha)}. \tag{58}$$

Solving for $xL_n^{(\alpha)}$,

$$xL_n^{(\alpha)} = (2n + \alpha + 1)L_n^{(\alpha)} - (n+1)L_{n+1}^{(\alpha)} - (n + \alpha)L_{n-1}^{(\alpha)},$$

so $A_n = -(n+1)$, $B_n = 2n + \alpha + 1$ and $C_n = -(n + \alpha)$. With $k_n = (-1)^n/n!$ we have $k_{n-1}/k_n = -n$ and $k_n/k_{n+1} = -(n+1) = A_n$, as the lemma requires. By (46) the monic coefficients are

$$b_n = 2n + \alpha + 1, \qquad a_n^2 = -(n + \alpha) \cdot (-n) = n(n + \alpha), \tag{59}$$

the two minus signs cancelling; this is why the monic coefficients are positive for $\alpha > -1$. The monic norm follows from (13):

$$h_n^{\mathrm{mon}} = h_0 \prod_{j=1}^{n} j(j + \alpha) = \Gamma(\alpha + 1)\, n!\, (\alpha + 1)_n = n!\, \Gamma(n + \alpha + 1),$$

and the norms in the standard normalisation are

$$h_n^{(\alpha)} = \frac{\Gamma(n + \alpha + 1)}{n!}, \qquad h_n^{\mathrm{mon}} = n!\, \Gamma(n + \alpha + 1). \tag{60}$$

The first polynomials are

$$L_0 = 1, \quad L_1^{(\alpha)} = \alpha + 1 - x, \quad L_2^{(\alpha)} = \frac{(\alpha+1)(\alpha+2)}{2} - (\alpha+2)x + \frac{x^2}{2}, \tag{61}$$

so that in the monic normalisation

$$Q_1 = x - \alpha - 1, \qquad Q_2 = x^2 - (2\alpha+4)x + (\alpha+1)(\alpha+2). \tag{62}$$

3.3 Hermite

The Hermite polynomials (physicists' normalisation) are orthogonal on $\mathbb{R}$ with weight e^{-x^2}. They satisfy

$$y'' - 2xy' + 2ny = 0, \tag{63}$$

the Rodrigues formula (44), have leading coefficient $k_n = 2^n$, and obey

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x). \tag{64}$$

Solving for xH_n,

$$xH_n = \frac{1}{2}H_{n+1} + nH_{n-1},$$

so $A_n = \frac{1}{2}$, $B_n = 0$ and $C_n = n$. With $k_n = 2^n$ we have $k_n/k_{n+1} = \frac{1}{2} = A_n$ and $k_{n-1}/k_n = \frac{1}{2}$, so (46) gives the monic coefficients

$$b_n = 0, \qquad a_n^2 = \frac{n}{2}. \tag{65}$$

The norms follow from $h_n = h_0 \prod_{j=1}^{n} a_j^2$ with $h_0 = \int_{\mathbb{R}} e^{-x^2}\mathrm{d}x = \sqrt{\pi}$:

$$h_n^{\mathrm{mon}} = \sqrt{\pi}\prod_{j=1}^{n}\frac{j}{2} = \frac{n!\sqrt{\pi}}{2^n}, \qquad h_n^{\mathrm{std}} = h_n^{\mathrm{mon}}k_n^2 = 2^n n!\sqrt{\pi}.$$

The first polynomials are

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x, \tag{66}$$

and their monic counterparts are

$$Q_0 = 1, \qquad Q_1 = x, \qquad Q_2 = x^2 - \frac{1}{2}, \qquad Q_3 = x^3 - \frac{3}{2}x. \tag{67}$$

They come out of the recurrence in one line each: $Q_2 = xQ_1 - a_1^2 Q_0 = x^2 - \frac{1}{2}$ and $Q_3 = xQ_2 - a_2^2 Q_1 = x^3 - \frac{1}{2}x - x$, where $a_1^2 = \frac{1}{2}$ and $a_2^2 = 1$. Note that $Q_2 = x^2 - \frac{1}{2}$, not $x^2 - 1$: the coefficient a_2^2 is $2/2 = 1$ only in the *tail* indexing of the associated family. This is the first place where the difference between the family and its associate shows up, and it is a useful check when reading §7.

3.4 The same data in monic normalisation

Table 2 collects the coefficients. Two features are worth pointing out because they are used constantly below.

First, the Jacobi coefficients simplify when written in the combination $2n+\alpha+\beta$; every denominator in (51)–(52) is a product of consecutive values of this quantity. Second, the Laguerre and Hermite coefficients are already polynomial in n, which is why their associated families are the ones with the simplest closed forms.

The norms in Table 2 are the ones that make (4) true; the identity is proved in §3.6.

3.5 The structure relation

The differential equation and the three-term recurrence are linked by a relation between Q_n and Q_{n-1} that involves a *shift of the parameters*. In the standard normalisations it reads

$$\frac{\mathrm{d}}{\mathrm{d}x}P_n^{(\alpha,\beta)}(x) = \frac{n+\alpha+\beta+1}{2}\,P_{n-1}^{(\alpha+1,\beta+1)}(x), \tag{68}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}L_n^{(\alpha)}(x) = -L_{n-1}^{(\alpha+1)}(x), \tag{69}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}H_n(x) = 2n\,H_{n-1}(x). \tag{70}$$

In the monic normalisation the three collapse into one formula.

Proposition 9. *Let $Q_n(\cdot\,; \text{parameters})$ be the monic classical polynomial of degree n. Then*

$$Q_n'(x; \text{parameters}) = n\,Q_{n-1}(x; \text{parameters}+1), \tag{71}$$

where $+1$ means $(\alpha,\beta)\to(\alpha+1,\beta+1)$ for Jacobi, $\alpha\to\alpha+1$ for Laguerre, and nothing for Hermite.

Proof. Jacobi. Since $Q_n = P_n/k_n$,

$$Q_n' = k_n^{-1}\frac{n+\alpha+\beta+1}{2}P_{n-1}^{(\alpha+1,\beta+1)} = \frac{n+\alpha+\beta+1}{2}\frac{k_{n-1}(\alpha+1,\beta+1)}{k_n}Q_{n-1}(\cdot;\alpha+1,\beta+1),$$

and the ratio of the leading coefficients is

$$\frac{k_{n-1}(\alpha+1,\beta+1)}{k_n} = \frac{(n+\alpha+\beta+1)_{n-1}/(2^{n-1}(n-1)!)}{(n+\alpha+\beta+1)_n/(2^n n!)} = \frac{2n}{n+\alpha+\beta+1},$$

so the factor is n.

Laguerre. With $Q_n = (-1)^n n! L_n^{(\alpha)}$,

$$Q_n' = (-1)^n n!\,(-1)L_{n-1}^{(\alpha+1)} = n\,(-1)^{n-1}(n-1)!L_{n-1}^{(\alpha+1)} = n\,Q_{n-1}(\cdot;\alpha+1).$$

Hermite. With $Q_n = 2^{-n}H_n$, $Q_n' = 2^{-n}\,2nH_{n-1} = n\,2^{-(n-1)}H_{n-1} = nQ_{n-1}$. □

The formula is worth memorising in the form (5): *differentiating the monic classical polynomial of degree n gives n times the monic polynomial of degree $n-1$ with every parameter increased by one.* It is the cleanest statement in the classical theory. It also explains why parameter shifts are everywhere in this subject: in the monic normalisation, differentiating and shifting the parameters are the same operation. It does *not* construct the associated family of §7; that family comes from shifting the *recurrence*, a different operation. The two are easy to confuse, and keeping them apart is the point of §7.

3.6 The norms are forced by the recurrence

Proposition 10. *For any monic family satisfying (3) with $h_n = \langle Q_n, Q_n\rangle$,*

$$h_n = a_n^2 h_{n-1} \qquad (n\ge 1). \tag{72}$$

Proof. Q_n and xQ_{n-1} are both monic of degree n, so their difference has degree at most $n-1$ and is orthogonal to Q_n:

$$h_n = \langle Q_n, Q_n\rangle = \langle Q_n, xQ_{n-1}\rangle = \langle xQ_n, Q_{n-1}\rangle.$$

By (3), $xQ_n = Q_{n+1} + b_nQ_n + a_n^2Q_{n-1}$; the first two terms are orthogonal to Q_{n-1}, so $\langle xQ_n, Q_{n-1}\rangle = a_n^2\langle Q_{n-1}, Q_{n-1}\rangle = a_n^2 h_{n-1}$. □

The proposition is the reason the norm formulas of Table 2 contain no free constants beyond h_0: the whole sequence of norms is the product $h_n = h_0\prod_{j=1}^{n} a_j^2$, and h_0 is the mass of the weight. In the standard normalisation the same identity reads

$$h_n^{\mathrm{std}} = a_n^2\Big(\frac{k_n}{k_{n-1}}\Big)^2 h_{n-1}^{\mathrm{std}}. \tag{73}$$

With the three families in hand—their equations, their recurrences, their norms—two questions arise: where do these families come from, and why are they the ones that govern so many physical problems? The next section answers both at once. Separation of variables produces exactly these three families, as the eigenfunctions of Laplace's equation, of the hydrogen atom and of the oscillator, and the mechanism is the same in all three cases.

4 The families in physics

The three families of §3 were not invented as exercises in orthogonality. Legendre introduced his polynomials in 1785 in the study of the attraction of spheroids, and they are still the coefficients of the multipole expansion of the electrostatic and gravitational potential; the Laguerre and Hermite families became working objects of physics in 1925–26, when the hydrogen atom and the harmonic oscillator were solved. This section explains where these appearances come from, and derives them from the equations of §3. The explanation is separation of variables, and it is short enough to be given in full: a partial differential equation with a symmetry, written in coordinates adapted to that symmetry, falls apart into one-variable equations, and those equations are the classical equations of §3.

One mechanism will be used three times, and it is worth naming before it is used. In each problem the one-variable equation has a series solution expanded about one end of its interval, and the series must terminate for the solution to be physically admissible: the potential must be finite on the sphere, the wave function must be normalizable. Termination happens exactly when a parameter of the equation equals a nonnegative integer. That parameter is the quantum number, and the degree of the terminating polynomial is the n of §3. The quantum numbers are therefore not labels attached to polynomials after the fact; they are the numbers forced by termination. Exercises 15–17 at the end practise the section.

4.1 Why separation of variables gives these polynomials

Let $u(q_1, \dots, q_N)$ satisfy a linear equation $Lu = 0$ that is unchanged by rotations and translations—Laplace's equation and the Schrödinger equations below are the examples—and use coordinates adapted to them. The product ansatz $u = f_1(q_1)\cdots f_N(q_N)$ turns $Lu = 0$ into N ordinary equations,

one for each factor, coupled only through the separation constants. The equation for f_j can always be put in the self-adjoint form

$$(pf_j')' + (\lambda_j w - r_j) f_j = 0 \tag{74}$$

on the interval of q_j. Here p, w, r_j are known functions of q_j produced by the change of variables: w is the Jacobian factor, so that $w\,\mathrm{d}q_j$ is the physical volume element in that coordinate; p is the coefficient left after multiplying the equation into self-adjoint form; and r_j collects the terms involving the other separation constants. Its eigenfunctions are orthogonal with weight w, by the argument of §2.5: multiply the equation for f_j by f_k and the equation for f_k by f_j, subtract, integrate and use integration by parts; the derivative terms cancel and what is left is

$$(\lambda_j - \lambda_k) \int f_j f_k \, w \, \mathrm{d}q_j = 0. \tag{75}$$

This is the orthogonality of §1, in the form in which physics produces it. It is a consequence of (74), not an extra assumption.

Two features of (74) explain the shape of Table 1, and both are elementary.

The weight is the Jacobian. The inner product of eigenfunctions is an integral over the physical region, so the weight is whatever the volume element produces. On the sphere in d dimensions the solid angle is $\mathrm{d}\Omega_{d-1} = \sin^{d-2}\theta \, \mathrm{d}\theta \, \mathrm{d}\Omega_{d-2}$, and with $x = \cos\theta$ this is

$$\mathrm{d}\Omega_{d-1} = (1-x^2)^{(d-3)/2} \, \mathrm{d}x \, \mathrm{d}\Omega_{d-2}. \tag{76}$$

The weight of the eigenfunctions of the polar angle is therefore $(1-x^2)^{(d-3)/2}$, which is the Jacobi weight with $\alpha = \beta = (d-3)/2$. At $d = 3$ it is 1, the Legendre weight; at $d = 2$ it is $(1-x^2)^{-1/2}$, the Chebyshev weight, which is the weight of the circle since $\mathrm{d}\varphi = \mathrm{d}x/\sqrt{1-x^2}$; at $d = 4$ it is $(1-x^2)^{1/2}$. The parameters α and β of Jacobi are not free decorations: they record the dimension of the space, and, once the remaining angles are separated, their quantum numbers.

The equation is the Bochner operator. The polar-angle part of the Laplacian, written in $x = \cos\theta$, is

$$\frac{1}{\sin^{d-2}\theta} \frac{\mathrm{d}}{\mathrm{d}\theta}\Big(\sin^{d-2}\theta \frac{\mathrm{d}}{\mathrm{d}\theta}\Big) = (1-x^2)\frac{\mathrm{d}^2}{\mathrm{d}x^2} - (d-1)x\frac{\mathrm{d}}{\mathrm{d}x}, \tag{77}$$

and the right-hand side is $\sigma \mathrm{d}^2/\mathrm{d}x^2 + \tau \mathrm{d}/\mathrm{d}x$ with $\sigma = 1-x^2$ and $\tau = -(d-1)x$: the polynomial coefficients of Definition 5. Bochner's theorem now applies to the whole problem at once, without any further computation. The sphere is symmetric under $x \mapsto -x$, so the two ends of the interval carry the same weight and $\alpha = \beta$; comparing $\tau = -(\alpha+\beta+2)x$ with $\tau = -(d-1)x$ gives $\alpha + \beta = d-3$, hence $\alpha = \beta = (d-3)/2$, in agreement with (76). The polynomial solutions are the Jacobi family, and the special cases are Legendre at $d = 3$ and the two families below. Everything in this section is a special case of this one computation.

The third feature is where the quantum numbers come from. A series solution about an endpoint has coefficients obeying $c_{k+2} = f(k)c_k$. The series terminates only if $f(k)$ vanishes for some k, and the vanishing condition has the form "parameter = integer". If the series does not terminate, the solution fails the physical requirement, and it fails in one of two ways: its coefficients decay too slowly for the series to converge at the endpoint (Legendre), or its coefficients grow so fast that the solution is exponentially large where it must not be (Laguerre and Hermite). In either case the admissible solutions are the terminating ones. They are polynomials, and their degrees are the quantum numbers.

4.2 Laplace's equation and Legendre

A static potential in a region free of charge satisfies Laplace's equation $\Delta V = 0$; where there is charge of density ρ it satisfies $\Delta V = -\rho/\varepsilon_0$, and the gravitational potential with mass density ρ satisfies $\Delta V = 4\pi G\rho$. Only the charge-free equation is needed for the angular problem, so nothing is lost by studying Laplace's equation. In spherical coordinates (r, θ, φ), with θ the polar angle,

$$\Delta V = \frac{1}{r^2}\frac{\partial}{\partial r}\Big(r^2\frac{\partial V}{\partial r}\Big) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\Big(\sin\theta\frac{\partial V}{\partial\theta}\Big) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 V}{\partial\varphi^2}. \tag{78}$$

Step 1: the azimuthal angle. Put $V = R(r)\Theta(\theta)\Phi(\varphi)$, multiply (78) by $r^2/(R\Theta\Phi)$ and multiply by $\sin^2\theta$. The result is

$$\sin^2\theta\,\frac{(r^2R')'}{R} + \frac{\sin\theta}{\Theta}\big(\sin\theta\,\Theta'\big)' + \frac{\Phi''}{\Phi} = 0. \tag{79}$$

The first two terms depend on r and θ, the third only on φ; so the third is a constant, which we call $-m^2$. Since φ and $\varphi + 2\pi$ are the same physical point, $\Phi(\varphi + 2\pi) = \Phi(\varphi)$, and therefore m is an integer:

$$\Phi'' = -m^2\Phi, \qquad m \in \mathbb{Z}. \tag{80}$$

Step 2: the radial and polar angles. With $\Phi''/\Phi = -m^2$, equation (79) separates again: divide by $\sin^2\theta$ and move the θ-term to the right. The left side depends only on r and the right only on θ, so both equal a constant. Writing the constant as $\ell(\ell+1)$,

$$(r^2R')' = \ell(\ell+1)R, \qquad \frac{1}{\sin\theta}\big(\sin\theta\,\Theta'\big)' + \Big[\ell(\ell+1) - \frac{m^2}{\sin^2\theta}\Big]\Theta = 0. \tag{81}$$

Every real number λ with $\lambda > -\frac{1}{4}$ can be written as $\ell(\ell+1)$ with real ℓ, so nothing is lost by this notation; it is chosen because the solutions below are polynomials in exactly this parametrisation.

Step 3: the radial equation. For $R = r^s$ one has $(r^2R')' = s(s+1)r^s$, so (81) holds for $s = \ell$ and for $s = -\ell - 1$, and these are the two independent solutions:

$$R(r) = Ar^{\ell} + Br^{-\ell-1}. \tag{82}$$

This is the origin of the statement that the ℓ-th multipole falls off like $r^{-\ell-1}$: the exponent is fixed by the radial equation.

Step 4: the polar angle. With $x = \cos\theta$ one has $\mathrm{d}/\mathrm{d}\theta = -\sin\theta\,\mathrm{d}/\mathrm{d}x$. Write $s = \sin\theta$ and $D = \mathrm{d}/\mathrm{d}x$. Then $\mathrm{d}/\mathrm{d}\theta = -sD$, so

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\Big(\sin\theta\frac{\mathrm{d}}{\mathrm{d}\theta}\Big) = \frac{1}{s}(-sD)\big(s(-sD)\big) = D\big(s^2D\big) = s^2\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{\mathrm{d}(s^2)}{\mathrm{d}x}\frac{\mathrm{d}}{\mathrm{d}x},$$

and $s^2 = 1 - x^2$ gives

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\Big(\sin\theta\frac{\mathrm{d}}{\mathrm{d}\theta}\Big) = (1-x^2)\frac{\mathrm{d}^2}{\mathrm{d}x^2} - 2x\frac{\mathrm{d}}{\mathrm{d}x}. \tag{83}$$

With $m = 0$, the second equation of (81) therefore reads

$$(1-x^2)\Theta'' - 2x\Theta' + \ell(\ell+1)\Theta = 0, \tag{84}$$

which is (47) with $\alpha = \beta = 0$: the Legendre equation of §3, with ℓ in the role of n. The operator in (83) *is* the Legendre operator, and this is why the Legendre equation, and not some other equation, governs the polar angle of every spherically symmetric problem.

Step 5: why ℓ is an integer. The interval is $-1 < x < 1$, and $x = \pm 1$ are the two poles of the sphere, which are ordinary points of space. A potential that is finite there must have Θ bounded at both ends. The following lemma, whose proof is the series computation that was postponed in §3, turns this into a condition on ℓ.

Lemma 11. *Let λ be real and let $y = \sum_{k\geq 0} c_k x^k$ solve $(1-x^2)y'' - 2xy' + \lambda y = 0$. Then*

$$c_{k+2} = \frac{k(k+1) - \lambda}{(k+1)(k+2)}\, c_k. \tag{85}$$

There is a polynomial solution if and only if $\lambda = \ell(\ell+1)$ for some $\ell \in \{0, 1, 2, \dots\}$, and then the polynomial solution has degree ℓ and is unique up to a constant. If λ is not of this form, every solution has $c_k \sim C/k$ with $C \neq 0$ along one parity class, and no solution is bounded at both ends of $(-1, 1)$.

Proof. Substituting the series in the equation and collecting the coefficient of x^k gives

$$(k+2)(k+1)c_{k+2} = \big[k(k-1) + 2k - \lambda\big]c_k = \big[k(k+1) - \lambda\big]c_k,$$

which is (85). Even and odd coefficients are computed independently from c_0 and c_1, so the solution space is spanned by one even and one odd series. A parity class terminates if and only if $k(k+1) = \lambda$ for some k of that parity; since $k(k+1)$ is strictly increasing in $k \geq 0$ within each parity, this happens for at most one k, and the condition is $\lambda = \ell(\ell+1)$. If it happens, the series is the polynomial of degree ℓ, determined up to c_0 or c_1. If it does not happen, then for large k

$$\frac{c_{k+2}}{c_k} = \frac{k^2 + k - \lambda}{k^2 + 3k + 2} = 1 - \frac{2}{k} + O\big(k^{-2}\big),$$

so kc_k tends to a nonzero constant along the parity class, i.e. $c_k \sim C/k$; the series at $x = 1$ then diverges like the harmonic series, and so the solution is unbounded at the pole. (The numerical values for $\lambda = \frac{3}{4}$ are $kc_k = -0.614, -0.602, -0.596, -0.594$ at $k = 10, 20, 40, 60$.) □

The conclusion is that ℓ is a nonnegative integer, so the polar angle carries the Legendre polynomial $P_\ell(\cos\theta)$, and the degree of the polynomial is the angular quantum number. The quantum number is a consequence of the geometry: finiteness at the two poles forces termination. The other parity class gives the second solution, which is excluded because it diverges at the poles.

Step 6: the solution. In the axisymmetric case $m = 0$ the general solution is a superposition of the two radial powers for every ℓ:

$$V(r, \theta) = \sum_{\ell=0}^{\infty} \big(A_\ell r^\ell + B_\ell r^{-\ell-1}\big) P_\ell(\cos\theta). \tag{86}$$

The coefficients are fixed by the boundary conditions: at the origin $r = 0$ only the r^ℓ terms are finite, at infinity only the $r^{-\ell-1}$ terms vanish, and on a surface where V is prescribed the coefficients are the coefficients of the expansion of the boundary data in Legendre polynomials. Because $P_\ell(\cos\theta)$

are orthogonal on $-1<x<1$ with weight 1, and $\int_{-1}^{1} P_\ell(x)^2\,\mathrm{d}x = 2/(2\ell+1)$, the coefficient of P_ℓ in the boundary data is obtained by the projection

$$\frac{2\ell+1}{2}\int_{-1}^{1} f(x)P_\ell(x)\,\mathrm{d}x, \tag{87}$$

which is the orthogonality of §3 used as a computational tool: the same formula as the Fourier coefficient, with the Legendre polynomials in place of the trigonometric functions.

Example 12 (A sphere at $V_0\cos^2\theta$)**.** *A conducting sphere of radius a is held at the potential $V_0\cos^2\theta$. Find the potential inside and outside.*

Inside. *Finiteness at the origin kills the B_ℓ, so $V=\sum A_\ell r^\ell P_\ell(\cos\theta)$. On the sphere, $V(a,\theta)=V_0\cos^2\theta$, and*

$$\cos^2\theta = \frac{1}{3}+\frac{2}{3}P_2(\cos\theta)$$

because $P_2=(3\cos^2\theta-1)/2$. Comparing coefficients, $A_0=\frac{1}{3}V_0$, $A_2a^2=\frac{2}{3}V_0$, and all other A_ℓ vanish:

$$V_{\text{in}}(r,\theta)=V_0\Big[\frac{1}{3}+\frac{2}{3}\Big(\frac{r}{a}\Big)^2P_2(\cos\theta)\Big].$$

Outside. *Vanishing at infinity kills the A_ℓ, so $V=\sum B_\ell r^{-\ell-1}P_\ell(\cos\theta)$. The same comparison on the sphere gives $B_0/a=\frac{1}{3}V_0$ and $B_2/a^3=\frac{2}{3}V_0$, hence*

$$V_{\text{out}}(r,\theta)=V_0\Big[\frac{a}{3r}+\frac{2}{3}\Big(\frac{a}{r}\Big)^3P_2(\cos\theta)\Big].$$

Both formulas reproduce $V_0\cos^2\theta$ at $r=a$, and they are different functions of r because the same boundary data are continued in the two directions. The outside potential is the potential of a point charge of magnitude $Q=4\pi\varepsilon_0 aV_0/3$ together with a quadrupole; the identification of the terms with charges is the subject of the next subsection.

4.3 The Coulomb kernel and the multipole expansion

The same computation, done for a single point charge, identifies the Legendre polynomials with a physical object: they are the coefficients in the expansion of $1/|\mathbf{r}-\mathbf{r}'|$. The identity behind the expansion is the generating function.

Proposition 13 (Generating function)**.** *For $|u|$ small enough that $|2xu-u^2|<1$,*

$$\frac{1}{\sqrt{1-2xu+u^2}}=\sum_{\ell=0}^{\infty}P_\ell(x)u^\ell. \tag{88}$$

Proof. Let $G(x,u)=(1-2xu+u^2)^{-1/2}=\sum c_\ell(x)u^\ell$ and differentiate: $(1-2xu+u^2)\partial_u G=(x-u)G$. Comparing the coefficient of u^ℓ gives

$$(\ell+1)c_{\ell+1}-2\ell x c_\ell+(\ell-1)c_{\ell-1}=xc_\ell-c_{\ell-1},$$

that is

$$(\ell+1)c_{\ell+1}=(2\ell+1)xc_\ell-\ell c_{\ell-1}, \tag{89}$$

with $c_0 = 1$ and $c_1 = x$. This is Bonnet's recurrence, which is the Legendre recurrence. A recurrence of this form determines all c_ℓ once c_0 and c_1 are fixed, and the Legendre polynomials satisfy the same recurrence with the same initial values $P_0 = 1$, $P_1 = x$ (their standard recursion is exactly (89) in this notation). Hence $c_\ell = P_\ell$ for every ℓ. □

Now put a point charge at the point $d\hat{\mathbf{z}}$ on the z-axis and evaluate the potential at $\mathbf{r} = (r, \theta, 0)$. The distance is $\sqrt{r^2 + d^2 - 2rd\cos\theta} = d\sqrt{1 - 2u\cos\theta + u^2}$ with $u = r/d$, so by (88)

$$\frac{1}{|\mathbf{r} - d\hat{\mathbf{z}}|} = \frac{1}{d}\sum_{\ell=0}^{\infty}\Big(\frac{r}{d}\Big)^{\ell} P_\ell(\cos\theta) \qquad (r < d). \tag{90}$$

The right-hand side is the potential of a point charge written as a sum over multipoles; each term is a solution of Laplace's equation, as it must be, because each is the product of a radial power (82) with an angular Legendre polynomial. For $r > d$ the same computation with r and d interchanged gives the same series with d/r in place of r/d.

For a general source point the distance depends on r, r' and the angle γ between the two position vectors, and rotation invariance allows the z-axis to be chosen along $\mathbf{r}'$. This gives the multipole kernel

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r_>}\sum_{\ell=0}^{\infty}\Big(\frac{r_<}{r_>}\Big)^{\ell} P_\ell(\cos\gamma), \qquad r_> = \max(r, r'), \quad r_< = \min(r, r'), \tag{91}$$

which is the single most useful formula in potential theory. For a charge distribution ρ contained in a bounded region, and for r larger than every r' in it, $r_> = r$ and $r_< = r'$, so

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{\ell=0}^{\infty}\frac{1}{r^{\ell+1}}\int \rho(\mathbf{r}')\, r'^{\ell} P_\ell(\cos\gamma)\,\mathrm{d}^3 r'. \tag{92}$$

The ℓ-th term is the ℓ-th *multipole*: a fixed function of the direction of $\mathbf{r}$, multiplied by $r^{-\ell-1}$. The first three terms are called the monopole, the dipole and the quadrupole:

$$\begin{gathered} \ell = 0: \ \frac{Q}{4\pi\varepsilon_0 r}; \qquad \ell = 1: \ \frac{\hat{\mathbf{r}}\cdot\mathbf{p}}{4\pi\varepsilon_0 r^2}; \qquad \ell = 2: \ \frac{Q_{ij}\hat{r}_i\hat{r}_j}{8\pi\varepsilon_0 r^3}, \\ Q = \int \rho\,\mathrm{d}^3 r', \qquad \mathbf{p} = \int \mathbf{r}'\rho\,\mathrm{d}^3 r', \qquad Q_{ij} = \int \big(3r'_i r'_j - r'^2\delta_{ij}\big)\rho\,\mathrm{d}^3 r', \end{gathered} \tag{93}$$

with a sum over $i, j = 1, 2, 3$ in the quadrupole term. The dipole formula follows from $P_1(\cos\gamma) = \cos\gamma = \hat{\mathbf{r}}\cdot\hat{\mathbf{r}}'$; the quadrupole formula from $r'^2 P_2(\cos\gamma) = \frac{1}{2}\big(3(\mathbf{r}'\cdot\hat{\mathbf{r}})^2 - r'^2\big)$. The series (92) converges for r outside the smallest sphere containing the source; inside the source the terms with $r_<$ must be kept, as in (90), and the expression is no longer a sum of multipoles.

Two consequences are worth recording. First, the multipoles of a distribution depend on the origin. A point charge at distance d from the origin has, about that origin, the moments qd^ℓ for every ℓ: the point charge is a monopole only about itself. The dipole term vanishes when the origin is the centre of charge (and, for gravity, the centre of mass), which is why the dipole is zero for a neutral atom and for the Earth. Second, the expansion is not a mathematical convenience but the physical reason for the selection rules of atomic and nuclear transitions: the interaction of an atom with an external field is dominated by the lowest nonvanishing multipole.

For gravity the only change is the replacement $1/(4\pi\varepsilon_0) \to -G$ with ρ the mass density; the same Legendre polynomials appear, the same $\ell = 1$ term vanishes about the centre of mass, and the $\ell = 2$

term is the flattening of a planet. For the Earth its coefficient in the standard normalisation is $J_2 = 1.08 \times 10^{-3}$, and it is what makes the orbital planes of satellites precess.

Finally, the multipole expansion is the large-r expansion of the Stieltjes transform of §6: (92) is the expansion of F_0 in powers of $1/z$ with $z = r$, and the convergents q_n/Q_n of §6 are the Padé approximants $[n-1/n]$ of that series. (A Padé approximant $[L/M]$ is the rational function with numerator degree at most L and denominator degree at most M whose expansion at infinity matches the series to the highest possible order.) The continued fraction is therefore an exact resummation of the multipole expansion, valid where the series is not; this is the reason the J-fraction appears in transport and scattering problems as well as in this one.

4.4 Azimuthal dependence and the spherical harmonics

When the boundary data depend on φ, $m \neq 0$ and the polar equation is the second equation of (81),

$$(1-x^2)\Theta'' - 2x\Theta' + \Big[\ell(\ell+1) - \frac{m^2}{1-x^2}\Big]\Theta = 0. \tag{94}$$

The extra term $m^2/(1-x^2)$ is singular at the poles, and it is this singularity that produces the associated Legendre functions. The solutions regular at $x = 1$ are obtained from the Legendre polynomials by differentiation. Differentiating (84) m times and multiplying by $(1-x^2)^{m/2}$ gives a solution of (94); explicitly,

$$P_\ell^m(x) = (1-x^2)^{m/2}\,\frac{\mathrm{d}^m}{\mathrm{d}x^m}P_\ell(x), \qquad 0 \le m \le \ell, \tag{95}$$

and for $m > \ell$ there is no solution regular at both poles, because the m-th derivative of a polynomial of degree ℓ vanishes. (The sign $(-1)^m$ used in the Condon–Shortley phase is a convention; we do not adopt it. For negative m one sets $P_\ell^{-m} = \frac{(\ell-m)!}{(\ell+m)!}P_\ell^m$.)

The associated Legendre functions are Jacobi polynomials. The leading coefficient (48) and the Rodrigues formula give

$$P_\ell^m(x) = \frac{(\ell+m)!}{2^m \ell!}\,(1-x^2)^{m/2} P_{\ell-m}^{(m,m)}(x), \tag{96}$$

as one checks at $(\ell, m) = (2, 1)$: $P_2 = \tfrac{1}{2}(3x^2-1)$, so $P_2^1 = 3x(1-x^2)^{1/2}$, while the right side is $\tfrac{6}{4}(1-x^2)^{1/2}P_1^{(1,1)} = \tfrac{3}{2}(1-x^2)^{1/2}\cdot 2x$, the same. Equivalently, the substitution $\Theta = (1-x^2)^{m/2}y$ carries (94) into

$$(1-x^2)y'' - 2(m+1)xy' + (\ell-m)(\ell+m+1)y = 0, \tag{97}$$

which is (47) with $\alpha = \beta = m$ and $n = \ell - m$, since $(\ell-m)(\ell+m+1) = n(n+2m+1)$. So the azimuthal quantum number m is the Jacobi parameter: $\alpha = \beta = |m|$. This is the second row of Table 3. (The substitution was checked symbolically for $m = 1, \dots, 4$ and $n \le 4$.)

The angular functions of the sphere are the spherical harmonics

$$Y_\ell^m(\theta,\varphi) = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}}\,P_\ell^m(\cos\theta)\,e^{im\varphi}, \qquad m = 0, \dots, \ell, \tag{98}$$

with $Y_\ell^{-m} = (-1)^m \overline{Y_\ell^m}$. They are orthonormal on the sphere,

$$\int_{S^2} Y_\ell^m\,\overline{Y_{\ell'}^{m'}}\,\mathrm{d}\Omega = \delta_{\ell\ell'}\delta_{mm'}, \tag{99}$$

which is (75) written with the weight 1 in $x = \cos\theta$ and with the φ-integral supplying $2\pi\delta_{mm'}$; the normalisation constant in (98) is the one that makes (99) hold. Every sufficiently smooth function on the sphere is a convergent sum of them, so the general solution of Laplace's equation is

$$V(r,\theta,\varphi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \big(A_{\ell m} r^{\ell} + B_{\ell m} r^{-\ell-1}\big) Y_\ell^m(\theta,\varphi), \tag{100}$$

and the boundary-value problem is solved exactly as before: expand the data in Y_ℓ^m, read off the coefficients, keep the powers that are finite in the region.

The spherical harmonics also convert (91) into the standard form of the multipole expansion. The addition theorem states that

$$P_\ell(\cos\gamma) = \frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} Y_\ell^m(\theta,\varphi)\, \overline{Y_\ell^m(\theta',\varphi')}, \tag{101}$$

where (θ,φ) and (θ',φ') are the directions of $\mathbf{r}$ and $\mathbf{r}'$. Inserting (101) in (92) gives

$$\begin{aligned} V(\mathbf{r}) &= \frac{1}{4\pi\varepsilon_0} \sum_{\ell=0}^{\infty} \frac{4\pi}{2\ell+1} \frac{1}{r^{\ell+1}} \sum_{m=-\ell}^{\ell} q_{\ell m} Y_\ell^m(\theta,\varphi), \\ q_{\ell m} &= \int \rho(\mathbf{r}')\, r'^{\ell}\, \overline{Y_\ell^m(\theta',\varphi')}\, \mathrm{d}^3 r', \end{aligned} \tag{102}$$

which is the form used in quantum mechanics: the numbers $q_{\ell m}$ are the multipole moments, and they transform under rotations as the $(2\ell+1)$-dimensional representation carried by Y_ℓ^m.

In d dimensions the standard name for the polar-angle family is Gegenbauer's. With $\lambda = (d-2)/2$,

$$C_\ell^{(\lambda)}(x) = \frac{(2\lambda)_\ell}{(\lambda+\frac12)_\ell} P_\ell^{(\lambda-1/2,\lambda-1/2)}(x), \tag{103}$$

so the Gegenbauer polynomials are the Jacobi family of (77) in another normalisation. At $d=3$ ($\lambda=\frac12$) the constant is 1 and $C_\ell^{(1/2)} = P_\ell$. At $d=2$ the family degenerates ($\lambda=0$), which is why the circle produces Chebyshev polynomials rather than a nondegenerate Gegenbauer family. Equation (103) was checked for $\ell \le 6$ and $\lambda = \frac12, 1, \frac32, \frac52, \frac72$; the Gegenbauer equation itself, $(1-x^2)y'' - (2\lambda+1)xy' + \ell(\ell+2\lambda)y = 0$, was checked on the same range.

4.5 The hydrogen atom and Laguerre

The hydrogen atom is the one problem in which two families appear at once. The electron of mass m_e and charge $-e$ moves in the Coulomb potential of a nucleus of charge Ze; the stationary states satisfy

$$-\frac{\hbar^2}{2m_e}\Delta\psi - \frac{Ze^2}{4\pi\varepsilon_0 r}\psi = E\psi. \tag{104}$$

Write $\psi = R(r) Y_\ell^m(\theta,\varphi)$. The angular part of the Laplacian acts on Y_ℓ^m as $-\ell(\ell+1)/r^2$ times Y_ℓ^m, which is the content of §4.2–§4.4, so the radial factor obeys

$$R'' + \frac{2}{r}R' + \Big[\frac{2m_e}{\hbar^2}\Big(E + \frac{Ze^2}{4\pi\varepsilon_0 r}\Big) - \frac{\ell(\ell+1)}{r^2}\Big] R = 0.$$

Put $u = rR$, so that $R'' + 2R'/r = u''/r$. This is the standard reduction of a radial equation to a one-dimensional one, and it gives

$$u'' + \Big[\frac{2m_e}{\hbar^2}\Big(E + \frac{Ze^2}{4\pi\varepsilon_0 r}\Big) - \frac{\ell(\ell+1)}{r^2}\Big]u = 0. \tag{105}$$

Equation (105) has the form of a one-dimensional Schrödinger equation with the effective potential $U(r) = -\frac{Ze^2}{4\pi\varepsilon_0 r} + \frac{\hbar^2\ell(\ell+1)}{2m_e r^2}$, the Coulomb well plus the centrifugal barrier. Bound states have $E < 0$; we look for them, and the condition $E < 0$ is what makes the equation solvable by polynomials.

For $E < 0$ set

$$\kappa = \frac{\sqrt{-2m_e E}}{\hbar}, \qquad \rho = 2\kappa r, \qquad \rho_0 = \frac{m_e Z e^2}{4\pi\varepsilon_0 \hbar^2 \kappa}. \tag{106}$$

Dividing (105) by $4\kappa^2$ and using $m_e E/(2\hbar^2\kappa^2) = -\frac{1}{4}$ turns it into

$$u'' + \Big[-\frac{1}{4} + \frac{\rho_0}{\rho} - \frac{\ell(\ell+1)}{\rho^2}\Big]u = 0, \tag{107}$$

where the primes are now derivatives in ρ. The behaviour at $\rho = 0$ is $u \sim \rho^{\ell+1}$ (the other exponent, $\rho^{-\ell}$, would make $R = u/r$ singular at the origin), and at $\rho = \infty$ it is $u \sim e^{-\rho/2}$ (the other behaviour, $e^{+\rho/2}$, is not normalizable). Removing both factors,

$$u = \rho^{\ell+1} e^{-\rho/2} v(\rho), \tag{108}$$

and a direct computation gives

$$\rho v'' + (2\ell + 2 - \rho)v' + (\rho_0 - \ell - 1)v = 0, \tag{109}$$

which is the Laguerre equation (57) with $\alpha = 2\ell + 1$ and $n = \rho_0 - \ell - 1$. The computation is worth seeing, because the coefficients $2\ell + 2$ and $\rho_0 - \ell - 1$ look arbitrary until they are traced back. Put $f = \rho^{\ell+1}e^{-\rho/2}$, so $u = fv$ and $u'' = f''v + 2f'v' + fv''$. From

$$\frac{f'}{f} = \frac{\ell+1}{\rho} - \frac{1}{2}, \qquad \frac{f''}{f} = \Big(\frac{f'}{f}\Big)' + \Big(\frac{f'}{f}\Big)^2 = \frac{\ell(\ell+1)}{\rho^2} - \frac{\ell+1}{\rho} + \frac{1}{4},$$

dividing (107) by f gives

$$v'' + 2\Big(\frac{\ell+1}{\rho} - \frac{1}{2}\Big)v' + \Big[\frac{\rho_0 - \ell - 1}{\rho}\Big]v = 0.$$

Multiplying by ρ gives (109): the $1/4$ cancels, the $\ell(\ell+1)/\rho^2$ cancels the centrifugal term, and the two surviving pieces are exactly the Laguerre coefficients. The parameter α is forced by the angular quantum number; it is not free.

It remains to see which v are admissible, and the mechanism is that of Lemma 11. The solution regular at $\rho = 0$ is the power series determined by (109); if the series does not terminate, its coefficients behave like $1/k!$ for large k, the series sums to a function growing like e^ρ, and $u = \rho^{\ell+1}e^{-\rho/2}v$ then grows like $e^{+\rho/2}$, which is not normalizable. So the series must terminate, and it terminates exactly when

$$\rho_0 - \ell - 1 = n_r, \qquad n_r = 0, 1, 2, \ldots, \tag{110}$$

in which case $v = L_{n_r}^{(2\ell+1)}$ is a Laguerre polynomial. Writing $n = n_r + \ell + 1$ (so that $n = 1, 2, \ldots$ and $\ell = 0, \ldots, n-1$), the condition $\rho_0 = n$ fixes the energy through (106):

$$E_n = -\frac{m_e}{2\hbar^2}\Big(\frac{Ze^2}{4\pi\varepsilon_0}\Big)^2 \frac{1}{n^2} = -13.6\,\frac{Z^2}{n^2}\ \text{eV}. \tag{111}$$

The radial functions are, up to normalisation,

$$R_{n\ell}(r) \propto \rho^{\ell} e^{-\rho/2} L_{n-\ell-1}^{(2\ell+1)}(\rho), \qquad \rho = \frac{2Zr}{na_0}, \tag{112}$$

where $a_0 = 4\pi\varepsilon_0\hbar^2/(m_e e^2) = 0.529\,\text{Å}$ is the Bohr radius and $\rho = 2\kappa r$ with $\kappa = Z/(na_0)$. The three quantum numbers are now all accounted for: ℓ and m from the angular equation, and n from the termination condition, hence the degree $n-\ell-1$ of the Laguerre polynomial, which is the number of radial nodes of the wave function and the number of zeros of $R_{n\ell}$ in $(0,\infty)$.

Three remarks. First, the eigenvalue problem for u is a Sturm–Liouville problem on $(0,\infty)$ with weight 1, so the radial functions for fixed ℓ are orthogonal in r; this is the physical orthogonality of the wave functions, $\int R_{n\ell} R_{n'\ell} r^2 \mathrm{d}r = \int u_{n\ell} u_{n'\ell} \mathrm{d}r$, and it is *not* the same statement as the Laguerre orthogonality of the polynomials, which is with weight $\rho^{2\ell+1}e^{-\rho}$ in the variable ρ—the two differ because the variable ρ contains the state-dependent κ. This is exactly the kind of conflation the list of confusions at the end warns against. Second, the energies (111) depend on n only, so states with different ℓ are degenerate; this degeneracy is larger than rotation invariance alone would allow, and its explanation (the conserved Runge–Lenz vector, an $SO(4)$ symmetry) is outside these notes, but the fact is visible in (111). Third, for fixed ℓ and large n the radial wave function oscillates $\sim n-\ell$ times, in agreement with the node count above.

The same Laguerre family describes the radial part of the three-dimensional isotropic oscillator, with the half-integer parameter $\alpha = \ell + \tfrac{1}{2}$: the radial equation for $V = \tfrac{1}{2}m_e\omega^2 r^2$ is solved by $R \propto r^{\ell} e^{-r^2/2} L_{n_r}^{(\ell+1/2)}(r^2)$ with $E = \hbar\omega(2n_r + \ell + \tfrac{3}{2})$. This was checked symbolically for $n_r \le 3$ and $\ell \le 3$. The half-integer parameter is not an accident, and it is the reason the Laguerre–Hermite limit of §5 is a physical statement: the same oscillator, written in Cartesian coordinates, is a product of three Hermite problems, and the two descriptions must agree.

4.6 The harmonic oscillator and Hermite

The one-dimensional oscillator, $-\frac{\hbar^2}{2m_e}\psi'' + \frac{1}{2}m_e\omega^2 x^2\psi = E\psi$, is made dimensionless by $\xi = x\sqrt{m_e\omega/\hbar}$ and $\varepsilon = 2E/(\hbar\omega)$:

$$\psi'' + (\varepsilon - \xi^2)\psi = 0. \tag{113}$$

At $\xi = \pm\infty$ the two behaviours are $e^{\pm\xi^2/2}$, and only $e^{-\xi^2/2}$ is normalizable, so put $\psi = e^{-\xi^2/2}H(\xi)$. Then

$$\psi' = e^{-\xi^2/2}(H' - \xi H), \qquad \psi'' = e^{-\xi^2/2}\big(H'' - 2\xi H' + (\xi^2 - 1)H\big),$$

and substituting in (113) cancels the ξ^2 terms and leaves

$$H'' - 2\xi H' + (\varepsilon - 1)H = 0, \tag{114}$$

which is the Hermite equation (63) with $\varepsilon - 1 = 2n$. The series solution about $\xi = 0$ has coefficients

$$c_{k+2} = \frac{2k - (\varepsilon - 1)}{(k+1)(k+2)}\,c_k, \tag{115}$$

as one sees by substituting $H = \sum c_k \xi^k$ and collecting the coefficient of ξ^k. The argument is the same as twice before: if the series does not terminate, $c_k \sim 1/k!$, the series sums to a function growing like e^{ξ^2}, and $\psi = e^{-\xi^2/2}H$ grows like $e^{+\xi^2/2}$, which is not normalizable. So the series terminates, which happens exactly when $\varepsilon - 1 = 2n$ with n a nonnegative integer, and

$$E_n = \hbar\omega\big(n + \tfrac{1}{2}\big), \qquad \psi_n(\xi) = N_n H_n(\xi) e^{-\xi^2/2}. \tag{116}$$

This was checked symbolically for $n \le 4$: ψ_n solves (113) with $\varepsilon = 2n + 1$.

The oscillator is also where the Jacobi matrix of §6 becomes a physical operator. Write the recurrence as $H_{n+1} = 2\xi H_n - 2nH_{n-1}$, or

$$\xi H_n = \tfrac{1}{2} H_{n+1} + n H_{n-1}. \tag{117}$$

In the orthonormal basis $\{\psi_n\}$ of (116) this says that multiplication by ξ has nonzero matrix elements only between n and $n \pm 1$. To see the values, write $\psi_n = N_n H_n(\xi) e^{-\xi^2/2}$. The standard norm $\int H_n^2 e^{-\xi^2}\,\mathrm{d}\xi = 2^n n!\sqrt{\pi}$ fixes $N_n = (2^n n!\sqrt{\pi})^{-1/2}$. Multiplying (117) by N_n, using $H_n = \psi_n/N_n$, and taking the inner product with ψ_m gives

$$\langle \psi_m | \xi | \psi_n \rangle = N_m N_n \Big[\tfrac{1}{2} \langle H_{n+1}, H_m \rangle + n \langle H_{n-1}, H_m \rangle \Big],$$

with the inner product weighted by $e^{-\xi^2}$. For $m = n + 1$ the bracket is $\frac{1}{2} 2^{n+1}(n+1)!\sqrt{\pi}$, and the prefactor reduces it to $\sqrt{(n+1)/2}$; for $m = n - 1$ the bracket is $n 2^{n-1}(n-1)!\sqrt{\pi}$, and the prefactor reduces it to $\sqrt{n/2}$. Hence

$$\langle \psi_m | \xi | \psi_n \rangle = \sqrt{\tfrac{n+1}{2}}\,\delta_{m,n+1} + \sqrt{\tfrac{n}{2}}\,\delta_{m,n-1}, \tag{118}$$

in oscillator units. This is the dipole selection rule $\Delta n = \pm 1$ of the oscillator, and it is nothing but the three-term recurrence read as a matrix identity. In general, for the orthonormal polynomials $\varphi_n = p_n/\sqrt{h_n^{\mathrm{std}}}$ of any weight, the matrix of multiplication by x is tridiagonal with diagonal b_n and off-diagonals a_{n+1}: it *is* the Jacobi matrix J of §6, in the basis $\{\varphi_n\}$. The general theory of §6 is therefore the spectral theory of multiplication by x, and for the oscillator that operator is the position operator of a quantum particle.

One more identification follows, and it closes the circle with §6. The Stieltjes transform (138) of the Hermite weight is the diagonal Green's function of the oscillator in its ground state:

$$F_0(z) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{z - t}\,\mathrm{d}t = \int_{-\infty}^{\infty} \frac{\psi_0(t)^2}{z - t}\,\mathrm{d}t = \langle \psi_0 | \frac{1}{z - \xi} | \psi_0 \rangle, \tag{119}$$

because $\psi_0^2 = e^{-t^2}/\sqrt{\pi}$. The continued fraction of §6 is the resolvent of the position operator, its convergents are its Padé approximants, and its spectral mass distribution is the ground-state density. Nothing in §6 was invented for this purpose, but this is where the objects of that section come from.

4.7 The cylinder does not give Bessel polynomials

Cylindrical coordinates separate Laplace's equation in the same way, and the radial equation is

$$\rho^2 R'' + \rho R' + (k^2\rho^2 - \nu^2) R = 0, \tag{120}$$

Bessel's equation, whose solutions are the cylindrical Bessel functions J_ν and Y_ν. These are not polynomials, and they are not the Bessel polynomials y_n of Table 1. The Bessel polynomials of

Bochner's classification are orthogonal on the unit circle, not on an interval of the real line, and they do not solve the radial problem of a cylinder; the fourth row of Table 1 therefore has no entry in Table 3. The separation of variables in a cylinder produces a transcendental family, and the same is true in spherical coordinates for the Helmholtz equation, where the radial factor is the spherical Bessel function $j_\ell(r) = \sqrt{\pi/2r}\, J_{\ell+1/2}(r)$, again not a polynomial. The polynomial families appear when the radial equation has a termination condition, which is what the Coulomb and oscillator potentials supply and a free particle does not.

4.8 The associated families in physics

The associated families of §7 are the recurrence with b_n and a_n^2 shifted by one in n, and they are not produced by the separation of variables of any local potential: the three families of the previous subsections exhaust what the standard problems give. They do arise in two other ways. First, in the tridiagonal representation approach, where the recurrence coefficients are the object of study and the association shift is the standard way of building a new exactly solvable model from a classical one [9]. Second, from a physical point of view, the shift is a shift of the angular momentum barrier: the associated family is the family whose eigenvalue is $(n+c)(n+c+1)$ rather than $n(n+1)$, which is what a centrifugal term does when the angular momentum is shifted by c. For integer c this is a relabelling; for non-integer c it is a fractional angular momentum, a situation that occurs in two-dimensional systems with anyons. The three associated weights written out in §7 are the spectral mass distributions of these shifted problems, and the fact that they are not of Pearson type (Proposition 26) is the statement that a shift of the angular momentum takes one outside the classical families.

The companion objects that *do* arise in physics are the second-kind functions of §6, not the associated polynomials. For Legendre, the second-kind function is

$$Q_\ell(x) = P_\ell(x) F_0(x) - k_\ell\, q_\ell(x), \qquad F_0(x) = \frac{1}{2} \ln \frac{1+x}{1-x}, \tag{121}$$

where k_ℓ is the leading coefficient of P_ℓ and q_ℓ is the second-kind polynomial of §6. At $\ell = 1$ this reads $Q_1(x) = xF_0(x) - 1$, which is the standard formula; at $\ell = 2$, $q_2 = x$ and $k_2 = \frac{3}{2}$, so $Q_2 = \frac{3}{2}\big[(x^2 - \frac{1}{3})F_0 - x\big]$, again the standard formula. So the polynomials q_n of §6 are the polynomial parts of the Legendre functions of the second kind, and the convergent identity (142) is the statement that the remainder in that formula is the tail of the continued fraction. The second-kind solution is the one singular on the axis, and it is the physical solution in regions where the axis is excluded, just as Y_ν is the physical solution outside a cylinder.

4.9 The families in one table

Table 3 can be read in either direction. Read downwards, it lists the four places where the classical families occur. Read across, it says what each family is: the polar angle of a sphere (Legendre), the polar angle of a sphere with azimuthal dependence or in d dimensions (Jacobi), the radial coordinate of a Coulomb or oscillator well (Laguerre), and a Cartesian coordinate of an oscillator (Hermite).

The one entry worth pausing on is the oscillator, which appears twice: once as Hermite, in Cartesian coordinates, and once as Laguerre with $\alpha = \ell + \frac{1}{2}$, in spherical coordinates. The two descriptions are of the same system, so the two families cannot be independent, and they are not: the limit of §5 that takes $\alpha \to \infty$ in Laguerre and produces Hermite is exactly the statement that the spherical

Family	Equation	Physical problem	Numbers
Legendre P_ℓ, Jacobi $(0,0)$	(47), $\alpha=\beta=0$	polar angle of Laplace's equation in three dimensions, $m=0$; multipole expansion of the potential	$\ell=0,1,2,\dots$
Jacobi $P_n^{(\alpha,\beta)}$	(47)	polar angle with azimuthal dependence, $\alpha=\beta=\lvert m\rvert$; polar angle in d dimensions, $\alpha=\beta=(d-3)/2$	$n=\ell-\lvert m\rvert$
Laguerre $L_n^{(\alpha)}$	(57)	radial hydrogen, $\alpha=2\ell+1$; radial isotropic oscillator, $\alpha=\ell+\frac{1}{2}$	$n=n_r$, ℓ
Hermite H_n	(63)	one-dimensional oscillator; one Cartesian direction of the isotropic oscillator	$n=0,1,2,\dots$
Chebyshev Jacobi $(-\frac{1}{2},-\frac{1}{2})$	(47), $\alpha=\beta=-\frac{1}{2}$	polar angle in two dimensions, i.e. on a circle	m

Table 3: The Bochner families in physics. The fourth family of Table 1, the Bessel polynomials, has no row: the cylinder produces the Bessel functions J_ν, which are not polynomials (§4.7). The weight in each row is the Jacobian (76).

and Cartesian descriptions agree. The half-integer values $\alpha=\frac{1}{2},\frac{3}{2},\frac{5}{2},\dots$ that the radial oscillator supplies are precisely the sequence along which this limit is taken, and the intermediate coefficient $a_n^2=n(n+\alpha)/(2\alpha)\to n/2$ in the proof of Proposition 16 is the algebraic form of the agreement. With this in hand, the algebraic limits of the next section are not a formal curiosity: they are the statement that the same physical problem can be solved in more than one coordinate system.

5 The limits between the families

The three families are not independent: Laguerre is a limit of Jacobi, and Hermite is a limit of Laguerre. The limits are cleanest in the monic normalisation, because no normalisation constants have to be tracked.

5.1 Jacobi to Laguerre

Proposition 14. *For fixed α and $z\in\mathbb{C}$,*

$$\lim_{\beta\to\infty}\beta^n\,Q_n^{(\alpha,\beta)}\Big(1-\frac{2z}{\beta}\Big)=(-2)^n\,Q_n^L(z;\alpha), \tag{122}$$

where $Q_n^L(z;\alpha)$ is the monic Laguerre polynomial.

Proof. Put $x=1-2w/\beta$ and define $P_n(w)=(-\beta/2)^nQ_n^{(\alpha,\beta)}(1-2w/\beta)$, a monic polynomial of degree n in w. Substituting in the monic recurrence and multiplying by $(-\beta/2)^n$ gives

$$wP_n=P_{n+1}+b'_nP_n+a'^2_nP_{n-1},\qquad b'_n=\frac{\beta(1-b_n)}{2},\quad a'^2_n=\frac{\beta^2a_n^2}{4}.$$

From (51), with $s=2n+\alpha$,

$$1-b_n=\frac{(s+\beta)(s+\beta+2)-\beta^2+\alpha^2}{(s+\beta)(s+\beta+2)}=\frac{(4n+2\alpha+2)\beta+O(1)}{\beta^2+O(\beta)},$$

so $b'_n \to 2n+\alpha+1$, the Laguerre coefficient. From (52),

$$a'^2_n = \frac{\beta^2}{4} \cdot \frac{4n(n+\alpha)(n+\beta)(n+\alpha+\beta)}{(2n+\alpha+\beta)^2(2n+\alpha+\beta+1)(2n+\alpha+\beta-1)} \longrightarrow n(n+\alpha),$$

the Laguerre coefficient as well. The recurrence coefficients converge, and the monic polynomial of degree n is a continuous function of the first n coefficients, each being a polynomial in those coefficients with integer coefficients; hence $P_n \to Q_n^L(\cdot;\alpha)$. Since $P_n = (-\beta/2)^n Q_n(1-2w/\beta)$, this is exactly (122). □

Remark 15. *The rate is $O(1/\beta)$, with constants that grow with n. For $n=1$ the limit can be checked by hand: with $Q_1^{(\alpha,\beta)}(x) = x + (\alpha-\beta)/(\alpha+\beta+2)$,*

$$\beta Q_1^{(\alpha,\beta)}\Big(1-\frac{2z}{\beta}\Big) = \beta - 2z + \frac{\beta(\alpha-\beta)}{\beta+\alpha+2} = 2\alpha + 2 - 2z + O(\beta^{-1}),$$

where $\beta(\alpha-\beta)/(\beta+\alpha+2) = \alpha-\beta+\alpha+2+O(\beta^{-1})$ was used. The right-hand side is $-2Q_1^L(z;\alpha)$.

Numerically, with $\alpha = 0.3$ and $z = 0.2+0.4\mathrm{i}$, the quantity $\left|\beta^n Q_n^{(\alpha,\beta)}(1-2z/\beta) - (-2)^n Q_n^L(z;\alpha)\right|$ is:

	$\beta = 10$	$\beta = 100$	$\beta = 1000$
$n=1$	2.4×10^{-1}	2.9×10^{-2}	3.0×10^{-3}
$n=2$	1.3	1.9×10^{-1}	2.0×10^{-2}
$n=3$	6.2	1.2	1.4×10^{-1}

The errors fall like $1/\beta$, which is the expected first-correction rate.

5.2 Laguerre to Hermite

Proposition 16. *For fixed $y \in \mathbb{C}$,*

$$\lim_{\alpha\to\infty} \alpha^{-n/2}\, Q_n^L\Big(\alpha + \sqrt{2\alpha}\, y; \alpha\Big) = 2^{n/2}\, Q_n^H(y), \tag{123}$$

where Q_n^H is the monic Hermite polynomial of (67).

Proof. Substitute $z = \alpha + \sqrt{2\alpha}\, y$ and put

$$P_n(y) = (2\alpha)^{-n/2} Q_n^L(z;\alpha).$$

This scaling is chosen to make P_n monic in y: the leading term of Q_n^L is $z^n = (\alpha+\sqrt{2\alpha}y)^n$, whose highest power is $(2\alpha)^{n/2}y^n$. In the Laguerre recurrence $zQ_n = Q_{n+1} + b_n Q_n + a_n^2 Q_{n-1}$ with $b_n = 2n+\alpha+1$, $a_n^2 = n(n+\alpha)$, write $Q_n = (2\alpha)^{n/2}P_n$. Dividing by $2^{n/2}\alpha^{n/2}\sqrt{2\alpha}$ gives the exact identity

$$yP_n = P_{n+1} + \frac{2n+1}{\sqrt{2\alpha}} P_n + \frac{n(n+\alpha)}{2\alpha} P_{n-1}. \tag{124}$$

Its coefficients tend to $b_n = 0$ and $a_n^2 = n/2$, the Hermite data, and the monic polynomial of degree n depends continuously on the first n recurrence coefficients. Hence $P_n \to Q_n^H$. Since $\alpha^{-n/2}Q_n^L = (2\alpha)^{n/2}\alpha^{-n/2}P_n = 2^{n/2}P_n$, this is exactly (123). □

The numerical errors, at $y = 0.6 + 0.3\mathrm{i}$, are:

	$\alpha = 10$	$\alpha = 100$	$\alpha = 1000$
$n = 1$	3.2×10^{-1}	1.0×10^{-1}	3.2×10^{-2}
$n = 2$	1.0	3.6×10^{-1}	1.2×10^{-1}
$n = 3$	2.4	6.8×10^{-1}	2.1×10^{-1}

They fall like $\alpha^{-1/2}$, one power slower than in the Jacobi–Laguerre limit, which is the signature of the square-root scaling.

5.3 The chain and the Askey table

The two limits compose. Taking first $\beta \to \infty$ and then $\alpha \to \infty$ in the Jacobi coefficients gives Hermite directly, so the three families sit at the bottom of a long chain

$$\text{(higher families)} \longrightarrow \text{Jacobi} \longrightarrow \text{Laguerre} \longrightarrow \text{Hermite}$$

whose upper reaches contain many further orthogonal polynomials, organised in the Askey scheme—a table of hypergeometric families that are related to one another by limits of this kind. The part of that chain above Jacobi amounts to taking further parameters to infinity in a four-parameter family (the Wilson polynomials, not treated here), and the part below Jacobi is exactly what Propositions 14 and 16 prove. Nothing above Jacobi is needed to understand the three families of these notes: each family is defined by its own weight, differential equation and recurrence, and is recovered from its neighbour by a limit of recurrence coefficients.

That last point is the one worth carrying away. Every limit in the chain is a statement about the numbers b_n and a_n^2: the polynomials follow because a monic polynomial is a continuous function of its recurrence coefficients. This is why each limit can be checked in a few lines on a computer, as the tables above do.

6 The Jacobi matrix, the Stieltjes transform, and the J-fraction

6.1 Spectral data

The three-term recurrence (3) can be read as the eigenvalue equation of a symmetric tridiagonal matrix. Let J be the infinite matrix with

$$J_{nn} = b_n, \qquad J_{n,n+1} = J_{n+1,n} = a_{n+1} \quad (a_{n+1} > 0), \tag{125}$$

and let $e_0 = (1, 0, 0, \dots)^T$. Concretely, J acts on a vector $v = (v_0, v_1, \dots)$ by

$$(Jv)_0 = b_0 v_0 + a_1 v_1, \qquad (Jv)_k = a_k v_{k-1} + b_k v_k + a_{k+1} v_{k+1} \quad (k \geq 1).$$

The passage from the recurrence to J rests on the following observations.

The finite sections. The expansion of a tridiagonal determinant along its last row is the computation of §1.8, and it gave (26), which is exactly the recurrence (3) with the index shifted. Hence

$$Q_n(x) = \det(xI - J_n), \tag{126}$$

where J_n is the $n \times n$ leading section of J. The monic orthogonal polynomials *are* the characteristic polynomials of the finite sections.

Real and simple zeros. Since J_n is real symmetric, its eigenvalues are real, and by (126) they are the zeros of Q_n. They are also simple. Indeed, if x is an eigenvalue with eigenvector $v \neq 0$, the first row of $(xI - J_n)v = 0$ reads $(x - b_0)v_0 - a_1 v_1 = 0$. If $v_0 = 0$ then $v_1 = 0$ because $a_1 > 0$, then $v_2 = 0$, and so on: $v = 0$, a contradiction. So every eigenvector has $v_0 \neq 0$, and two independent eigenvectors for the same x would combine to one with first component 0. Each eigenvalue therefore has a one-dimensional eigenspace, and Q_n has n distinct real zeros. (Setting x equal to a zero of Q_n and solving the recurrence (3) forwards from $v_0 = 1$ reproduces the eigenvector.)

Because J_n is obtained from J_{n-1} by adding one row and one column, the new eigenvalue falls between consecutive old ones. The elementary proof uses only signs. Let $y_1 < \cdots < y_{n-1}$ be the zeros of Q_{n-1}. At a zero of Q_{n-1}, the recurrence (26) gives

$$Q_n(y_j) = -a_{n-1}^2 Q_{n-2}(y_j).$$

By induction, the zeros $z_1 < \cdots < z_{n-2}$ of Q_{n-2} interlace those of Q_{n-1}: $y_1 < z_1 < y_2 < \cdots < z_{n-2} < y_{n-1}$. Hence $Q_{n-2}(y_j)$ changes sign when j increases by one; also $Q_{n-2}(y_1)$ has the sign of $Q_{n-2}(-\infty)$, and $Q_{n-2}(y_{n-1})$ the sign of $Q_{n-2}(+\infty)$. Therefore $Q_n(y_j)$ changes sign, and the intermediate value theorem gives one zero of Q_n in each interval (y_j, y_{j+1}). At the two ends, $Q_n(-\infty)$ has the opposite sign to $Q_n(y_1)$, and $Q_n(y_{n-1})$ has the opposite sign to $Q_n(+\infty)$, so there is one further zero below y_1 and one above y_{n-1}. This accounts for all n zeros: the zeros of Q_n interlace with those of Q_{n-1}.

The formal eigenvector. Put

$$v_k(x) = \frac{Q_k(x)}{a_1 a_2 \cdots a_k} \qquad (k \geq 0),$$

with the empty product equal to 1, so $v_0 = Q_0 = 1$. Then

$$(Jv)_k = \frac{Q_{k+1} + b_k Q_k + a_k^2 Q_{k-1}}{a_1 \cdots a_k} = \frac{x Q_k}{a_1 \cdots a_k} = x\, v_k,$$

where the middle step uses (3). Thus the orthogonal polynomials are the successive components of a solution of $(J - x)v = 0$: the recurrence (3) is a difference equation, and Q_n is the polynomial that the initial condition $v_0 = 1$, $v_{-1} = 0$ produces. (The second-kind polynomials of §6 correspond to the other natural initial condition.)

From finite sections to a mass distribution. A *mass distribution* is a rule that assigns a total mass to every interval; the two kinds used here are a weight, $\mathrm{d}\nu(t) = w(t)\,\mathrm{d}t$, and a finite sum of point masses. For a point mass δ_x at x, the integral is evaluation, $\int f(t)\,\mathrm{d}\delta_x(t) = f(x)$; for a finite sum, integrals are sums. With this notation, an infinite matrix is not a matrix in the first-year sense, and we never invert one. What we use is the sequence of finite matrices J_n. Each is real symmetric; let $x_{n,1} < \cdots < x_{n,n}$ be its eigenvalues and $u_{n,1}, \ldots, u_{n,n}$ an orthonormal basis of eigenvectors. The *spectral mass distribution of J_n at e_0* puts mass $|(e_0, u_{n,j})|^2$ at the point $x_{n,j}$ and no mass elsewhere. The masses add to 1 because the eigenvectors are an orthonormal basis. Its moments are

$$\int t^k\,\mathrm{d}\nu_n(t) = \sum_{j=1}^{n} x_{n,j}^k\,|(e_0, u_{n,j})|^2 = (e_0, J_n^k e_0),$$

the last equality being the eigenvector expansion of J_n^k. For fixed k and $n > k$ the right-hand side is independent of n, by the argument of §1.8; call the common value m_k. The *spectral mass distribution*

of the family is the mass distribution ν whose moments are the m_k:

$$\int t^k \,\mathrm{d}\nu(t) = m_k = (e_0, J_n^k e_0) \qquad (n > k). \tag{127}$$

We do not need a general existence theorem. In each case below ν is exhibited explicitly, and its moments are checked to be the m_k.

The classical mass distribution is the weight. For the three families of Table 1, take

$$\nu = \frac{w(x)\,\mathrm{d}x}{h_0}, \qquad h_0 = \int w\,\mathrm{d}x.$$

Its total mass is one, and its moments are m_k. This is checked by moments, and the check is short enough to do here. Since $Q_k(J_n)e_0 = a_1 a_2 \cdots a_k\, e_k$ (the recurrence (3) read in J_n), the expansion of the monic polynomial $t^k = Q_k(t) + c_{k-1}Q_{k-1}(t) + \cdots + c_0$ gives

$$J_n^k e_0 = Q_k(J_n)e_0 + c_{k-1}Q_{k-1}(J_n)e_0 + \cdots + c_0 e_0,$$

whose e_0 entry is c_0. On the other hand, integrating the same expansion against w and using $\int Q_j w\,\mathrm{d}x = 0$ for $j \geq 1$ and $\int Q_0 w\,\mathrm{d}x = h_0$ gives $\frac{1}{h_0}\int t^k w\,\mathrm{d}x = c_0$ as well. So the moments agree. For example,

$$(e_0, J_n e_0) = b_0, \qquad (e_0, J_n^2 e_0) = b_0^2 + a_1^2,$$

and for Legendre ($b_0 = 0$, $a_1^2 = \frac{1}{3}$) this is $\frac{1}{3} = \frac{1}{2}\int_{-1}^{1} t^2\,\mathrm{d}t$, while for Hermite ($b_0 = 0$, $a_1^2 = \frac{1}{2}$) it is $\frac{1}{2} = \frac{1}{\sqrt{\pi}}\int t^2 e^{-t^2}\,\mathrm{d}t$. This verifies the moment identity (127) for the three families, with no uniqueness theorem required: ν has been defined to be $w\,\mathrm{d}x/h_0$.

6.2 The Stieltjes transform

The *Stieltjes* (or *Cauchy*) transform of the mass distribution ν of (127) is the function

$$F_0(z) = \int_{\mathbb{R}} \frac{\mathrm{d}\nu(t)}{z - t}, \qquad z \in \mathbb{C} \setminus \mathbb{R}. \tag{128}$$

Because ν has total mass 1 and the singularity at $t = z$ is never reached, the integral converges for every z off the real axis, and differentiation under the integral sign (§1.6) shows that F_0 is differentiable to all orders on each half-plane, with

$$F_0'(z) = -\int_{\mathbb{R}} \frac{\mathrm{d}\nu(t)}{(z-t)^2}.$$

Moreover

$$F_0(z) = \frac{1}{z}\int \frac{\mathrm{d}\nu(t)}{1 - t/z} = \frac{1}{z} + O\Big(\frac{1}{z^2}\Big) \qquad (z \to \infty), \tag{129}$$

the term $O(z^{-2})$ being bounded by $\frac{1}{|z|^2}\int |t|\;\mathrm{d}\nu(t)/(1 - |t|\,/\,|z|)$, which is finite for large $|z|$.

The imaginary part of F_0 recovers the mass distribution, and this is the one computation that drives the whole of §6 and §7. Let $z = x + \mathrm{i}y$ with $y > 0$. Multiplying numerator and denominator by $x - t - \mathrm{i}y$,

$$\frac{1}{z - t} = \frac{x - t - \mathrm{i}y}{(x-t)^2 + y^2}, \tag{130}$$

so

$$\operatorname{Im} F_0(x+\mathrm{i}y) = -\int_{\mathbb{R}} \frac{y\,\mathrm{d}\nu(t)}{(x-t)^2+y^2} = -\int_{\mathbb{R}} \frac{y}{(x-t)^2+y^2}\,\nu'(t)\,\mathrm{d}t \tag{131}$$

whenever ν has a density ν'. Now apply the approximate-identity lemma (23) with $\phi = \nu'$: as $y \downarrow 0$ the integral tends to $\pi\nu'(x)$. Hence

$$\operatorname{Im} F_0(x+\mathrm{i}0) = -\pi\,\nu'(x) \tag{132}$$

at every point where ν has a continuous density ν'. In particular $\operatorname{Im} F_0 < 0$ in the upper half-plane, as (131) also shows directly.

Equation (132) is the bridge between the transform, which is what the continued fraction computes, and the weight, which is what the orthogonal polynomials integrate against. It is also the reason that every formula in §7 for an associated weight is a statement about $|F_{c-1}|$ on the real axis: taking imaginary parts of a relation between transforms converts the algebra of continued fractions into the analysis of weights.

6.3 What the J-fraction computes

Proposition 17. *The Stieltjes transform of the spectral mass distribution of the family (defined in §6) is the value of the infinite continued fraction*

$$F_0(z) = \cfrac{1}{z - b_0 - \cfrac{a_1^2}{z - b_1 - \cfrac{a_2^2}{z - b_2 - \cdots}}}, \tag{133}$$

and, more generally, the tail matrices $J^{(c)}$ (the submatrix on indices $c, c+1, \dots$) have transforms F_c satisfying

$$F_{c-1}(z) = \frac{1}{z - b_{c-1} - a_c^2 F_c(z)}, \qquad \text{i.e.} \qquad F_c(z) = \frac{z - b_{c-1} - 1/F_{c-1}(z)}{a_c^2}. \tag{134}$$

Proof. Everything is done in finite sections, whose inverses exist. Fix n, let J_n be the leading $n \times n$ section (for $n = 1$ the fraction is simply $1/(z - b_0)$), and put $u = (zI - J_n)^{-1} e_0$. By the eigenvector expansion that defines ν_n in §6,

$$u_0 = \sum_{j=1}^{n} \frac{|(e_0, u_{n,j})|^2}{z - x_{n,j}} = \int \frac{\mathrm{d}\nu_n(t)}{z-t} =: F_0^{(n)}(z).$$

The system $(zI - J_n)u = e_0$ reads, row by row,

$$(z-b_0)u_0 - a_1 u_1 = 1, \qquad -a_k u_{k-1} + (z-b_k)u_k - a_{k+1}u_{k+1} = 0 \quad (1 \le k \le n-2),$$

with last row $-a_{n-1}u_{n-2} + (z - b_{n-1})u_{n-1} = 0$. Set $w_j = u_{j+1}$, so $w_0 = u_1, \dots, w_{n-2} = u_{n-1}$. For $1 \le j \le n-2$ row $k = j+1$ reads

$$(z - b_{j+1})w_j - a_{j+2}w_{j+1} - a_{j+1}w_{j-1} = 0,$$

which is row j of the tail system; row $k=1$ is the same row with a right-hand side, $(z-b_1)w_0-a_2w_1 = a_1u_0$. In matrix form,

$$(zI - J_n^{(1)})w = a_1u_0\, e_0,$$

where $J_n^{(1)}$ is the $(n-1)\times(n-1)$ tail section. Hence

$$u_1 = w_0 = a_1u_0\, F_1^{(n)},$$

with $F_1^{(n)}$ the transform of the tail section. Substituting into the first row,

$$(z-b_0)u_0 - a_1^2u_0F_1^{(n)} = 1 \quad\Longrightarrow\quad F_0^{(n)} = \frac{1}{z-b_0-a_1^2F_1^{(n)}}.$$

The same argument applied to the tail gives $F_1^{(n)} = 1/(z-b_1-a_2^2F_2^{(n)})$, and so on down to $F_{n-1}^{(n)} = 1/(z-b_{n-1})$. Thus $F_0^{(n)}$ is exactly the nth convergent of (133).

It remains to pass to the limit. The infinite continued fraction is defined as the limit of its convergents, so the fraction computes $\lim_n F_0^{(n)}$. That limit is F_0: by §6 the mass distributions ν_n have the moments $m_0,\dots,m_{n-1}$ of ν, and for fixed z off the real axis the function $t\mapsto 1/(z-t)$ is continuous on the interval of the weight, which carries all the mass and contains all the $x_{n,j}$ (the last point is proved in §8). Weierstrass' theorem (a continuous function on a closed bounded interval is a uniform limit of polynomials) supplies a polynomial P with $|1/(z-t)-P(t)|\le\varepsilon$ there; then $\int P\,\mathrm{d}\nu_n = \int P\,\mathrm{d}\nu$ as soon as $n>\deg P$, so

$$\left|F_0^{(n)}(z) - F_0(z)\right| \le 2\varepsilon + \left|\int P\,(\mathrm{d}\nu_n - \mathrm{d}\nu)\right| = 2\varepsilon.$$

Hence $F_0^{(n)}\to F_0$ and (133) holds. Applying the same reasoning to the tails gives the nesting relation (134). □

A warning about normalisation is in order, because it is the one point where the algebra of this section is easy to get wrong. The continued fraction (133) represents the transform of the normalised mass distribution $\nu = w\,\mathrm{d}x/h_0$, of total mass 1, not the transform of $w\,\mathrm{d}x$ itself. If $F_0^w(z)=\int w(t)\,\mathrm{d}t/(z-t)$ is the transform of the weight as written in Table 1, then

$$F_0(z) = \frac{F_0^w(z)}{h_0}, \qquad h_0 = \int w\,\mathrm{d}x. \tag{135}$$

For Legendre with $w=1$ this gives $h_0=2$ and $F_0=\frac12\ln\frac{z+1}{z-1}$; for Hermite $h_0=\sqrt{\pi}$ and the transform is the familiar one divided by $\sqrt{\pi}$. The numerically verified statement is the one in (135); a quick check with Hermite at $z=0.5+2\mathrm{i}$ gives the continued fraction converging to $F_0^w(z)/\sqrt{\pi}$ to twelve digits.

6.4 Closed forms for the three families

For the three families the transform can be computed in closed form by elementary integrals. We give the transform of the normalised mass distribution (135).

Proposition 18. *In the upper half-plane,*

$$\text{Legendre } (w=1,\ h_0=2): \qquad F_0(z)=\frac{1}{2}\ln\frac{z+1}{z-1}, \tag{136}$$

$$\text{Laguerre } \alpha=0\ (w=e^{-x},\ h_0=1): \qquad F_0(z)=e^{-z}\big(\mathrm{Ei}(z)-\mathrm{i}\pi\big), \tag{137}$$

$$\text{Hermite } (w=e^{-x^2},\ h_0=\sqrt{\pi}): \qquad F_0(z)=-\mathrm{i}\sqrt{\pi}\,e^{-z^2}\operatorname{erfc}(-\mathrm{i}z). \tag{138}$$

Verification. Each formula is the statement that the right-hand side is the integral $\frac{1}{h_0}\int w(t)/(z-t)\,\mathrm{d}t$. For Legendre, $\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}z}\ln\frac{z+1}{z-1}=\frac{1}{2}\big(\frac{1}{z+1}-\frac{1}{z-1}\big)=-\frac{1}{z^2-1}$, and $\frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{2}\int_{-1}^{1}\frac{\mathrm{d}t}{z-t}=\frac{1}{2}\int_{-1}^{1}\frac{-\mathrm{d}t}{(z-t)^2}=-\frac{1}{2}\big(\frac{1}{z-1}-\frac{1}{z+1}\big)=-\frac{1}{z^2-1}$; both sides vanish as $z\to\infty$. For Laguerre, put $H(z)=\int_0^\infty e^{-t}/(z-t)\,\mathrm{d}t$. Differentiating under the integral and integrating by parts,

$$H'(z)=-\int_0^\infty\frac{e^{-t}}{(z-t)^2}\,\mathrm{d}t=\frac{1}{z}-H(z) \quad\Longrightarrow\quad H'+H=\frac{1}{z}.$$

The function $G(z)=e^{-z}(\mathrm{Ei}(z)-\mathrm{i}\pi)$ satisfies the same equation, $G'+G=e^{-z}\mathrm{Ei}'(z)=e^{-z}e^{z}/z=1/z$, and both H and G tend to 0 as $z\to\infty$ in the upper half-plane. Their difference $D=H-G$ therefore solves $D'+D=0$ with $D\to0$, so $D=0$. This proves the Laguerre formula and avoids introducing a second special function. For Hermite, $\int_{\mathbb{R}}e^{-t^2}/(z-t)\,\mathrm{d}t=-\mathrm{i}\pi e^{-z^2}\operatorname{erfc}(-\mathrm{i}z)$ is the standard representation of the complementary error function; dividing by $h_0=\sqrt{\pi}$ gives (138). □

All three formulas were checked against numerical quadrature at $z=0.4+0.7\mathrm{i}$: the quadrature values agree with (136)–(138) to fourteen digits, the differences being at the level of roundoff.

6.5 The second-kind polynomials and convergents

The continued fraction (133) has convergents that are ratios of polynomials of the second kind.

Definition 19. *The* second-kind polynomials q_n *of a monic family are defined by*

$$q_0=0, \qquad q_1=1, \qquad q_{n+1}(x)=(x-b_n)q_n(x)-a_n^2q_{n-1}(x). \tag{139}$$

The sequence q_n satisfies the same recurrence as Q_n but starts from the other natural initial condition, $q_0=0$, $q_1=1$; it is a polynomial of degree $n-1$. The two sequences are a basis of the solution space of the recurrence, and this is why every solution of the difference equation is a combination of Q_n and q_n.

An infinite continued fraction is defined as the limit of its truncations, so it is worth seeing the first two. Truncating (133) after one and after two levels,

$$\frac{1}{z-b_0}=\frac{q_1}{Q_1}, \qquad \frac{1}{z-b_0-\dfrac{a_1^2}{z-b_1}}=\frac{z-b_1}{(z-b_0)(z-b_1)-a_1^2}=\frac{q_2}{Q_2}, \tag{140}$$

which is the pattern: the nth convergent of (133) is q_n/Q_n. The general statement follows by multiplying the finite fraction out from the bottom, which reproduces (139) in the numerator and (3) in the denominator. The following identities make the convergence quantitative.

Proposition 20. *For $n \geq 1$,*

$$Q_n(z)q_{n-1}(z) - Q_{n-1}(z)q_n(z) = -\prod_{j=1}^{n-1} a_j^2, \tag{141}$$

$$Q_n(z)F_0(z) - q_n(z) = \Big(\prod_{j=1}^{n} a_j^2\Big)F_0(z)F_1(z)\cdots F_n(z), \tag{142}$$

where the empty product is 1. *Consequently, for each fixed n, the nth convergent agrees with F_0 to order z^{-2n-1} as $z \to \infty$.*

Verification. The determinant identity. Write $W_n = Q_nq_{n-1} - Q_{n-1}q_n$. Using (3) and (139),

$$Q_{n+1}q_n - Q_nq_{n+1} = \big[(x-b_n)Q_n - a_n^2Q_{n-1}\big]q_n - Q_n\big[(x-b_n)q_n - a_n^2q_{n-1}\big] = a_n^2W_n.$$

Since $W_1 = Q_1q_0 - Q_0q_1 = -1$, induction gives $W_n = -\prod_{j=1}^{n-1} a_j^2$.

The convergent identity. Write $C_n = Q_nF_0 - q_n$. The case $n = 0$ is $C_0 = F_0$, which is (142) with the empty product. For $n = 1$, (134) rearranged gives $C_1 = a_1^2F_0F_1$. For $n \geq 1$, subtracting the recurrence for q from the recurrence for Q multiplied by F_0 gives

$$C_{n+1} = (x-b_n)C_n - a_n^2C_{n-1}.$$

Substituting the claimed formula at levels n and $n-1$,

$$C_{n+1} = a_n^2\Big(\prod_{j=1}^{n-1} a_j^2\Big)F_0\cdots F_{n-1}\big[(x-b_n)F_n - 1\big],$$

and by (134) at level n the bracket is $a_{n+1}^2F_nF_{n+1}$. Hence

$$C_{n+1} = \Big(\prod_{j=1}^{n+1} a_j^2\Big)F_0\cdots F_{n+1},$$

which is the induction step.

Order of accuracy. By (129), $F_j(z) = 1/z + O(z^{-2})$ for every j, so the right-hand side of (142) is $\big(\prod_{j=1}^{n} a_j^2\big)z^{-n-1}(1 + O(1/z))$. Dividing by $Q_n(z) = z^n(1 + O(1/z))$ gives

$$F_0(z) - \frac{q_n}{Q_n} = O(z^{-2n-1}) \qquad (z \to \infty),$$

with a constant that depends on n but not on z. □

That the convergents converge as $n \to \infty$ for each fixed $z \in \mathbb{C} \setminus \mathbb{R}$, and not merely that each one matches F_0 far out in the plane, is the convergence theorem for J-fractions (Stieltjes); it holds for the three families here, whose coefficients grow at most linearly. We quote this once and use it only as a limit statement: the partial fractions of a finite symmetric matrix converge, and their limit is F_0. The following table shows the rate for Hermite at a point one unit off the real axis.

The identities were checked numerically for Hermite with $n \leq 4$ at $z = 0.4 + 0.7\mathrm{i}$; the residual of (142) is below 3×10^{-15}, the level of double precision. The convergence of the convergents is slow near the real axis, as it must be, and is illustrated by the following values of $|q_n/Q_n - F_0|$ at $z = 0.4 + 0.7\mathrm{i}$:

n	1	2	3	5	10	20	40
error	4.1×10^{-1}	2.1×10^{-1}	1.2×10^{-1}	4.7×10^{-2}	8.0×10^{-3}	6.3×10^{-4}	1.7×10^{-5}

7 The associated case

7.1 Definition

Definition 21. *Let $\{Q_n\}$ be a monic family with recurrence coefficients $\{b_n, a_n^2\}$. For a real number $c \geq 0$ the* associated family of order c *is the sequence $\{Q_n^{(c)}\}$ defined by*

$$Q_0^{(c)} = 1, \qquad Q_1^{(c)} = x - b_c, \qquad Q_{n+1}^{(c)} = (x - b_{n+c})Q_n^{(c)} - a_{n+c}^2 Q_{n-1}^{(c)}. \tag{143}$$

For $c = 0$ this is the original family. For $c = 1, 2, \ldots$ it is the family whose recurrence is the *tail* of the original one; this is the case for which the spectral picture is immediate. For non-integer c the right-hand side of (143) is still well defined, because b_{n+c} and a_{n+c}^2 are polynomial (or rational) functions of the index for the three classical families, and the resulting sequence is again a family of orthogonal polynomials provided the parameter range is respected (for instance $\alpha + c > -1$ for Laguerre).

7.2 The tail Jacobi matrix

Let $J^{(c)}$ be the infinite submatrix of J obtained by deleting the first c rows and columns, with entries b_{n+c} on the diagonal and a_{n+c+1} off the diagonal. It is symmetric and tridiagonal with positive off-diagonal entries, so every finite section is real symmetric and has real eigenvalues and an orthonormal basis of eigenvectors. Applying the construction of §6 to these finite sections gives a mass distribution ν_c of total mass 1 at the first vector e_0. The monic orthogonal polynomials of ν_c satisfy exactly (143): this is the same eigenvalue identity as in §6, applied to the tail.

Proposition 22. *For $c = 0, 1, 2, \ldots$ the family $\{Q_n^{(c)}\}$ is orthogonal with respect to a positive mass distribution ν_c of total mass 1, the spectral mass distribution of $J^{(c)}$. Its norms satisfy*

$$h_n^{(c)} = a_{n+c}^2 \, h_{n-1}^{(c)} \qquad (n \geq 1), \tag{144}$$

and the norms are determined by $h_0^{(c)} = 1$ and $h_n^{(c)} = \prod_{j=1}^{n} a_{j+c}^2$.

Proof. The first sentence is the construction of §6 applied to the tail: each finite section has a spectral mass distribution of total mass 1, its masses are squares and hence positive, and the limit distribution is ν_c. For non-integer c the same recurrence coefficients are used; the weight exhibited in §7 below shows that the resulting family is orthogonal in the stated parameter range. The norm identity is Proposition 10 applied to the tail recurrence, and the product formula is its iteration. □

Note that the norm identity (144) involves a_{n+c}^2, not a_n^2: the norms are those of the tail, as they must be. For the three classical families the products are elementary:

$$\begin{aligned}
\text{Jacobi: } h_n^{(c)} &= \prod_{j=1}^{n} \frac{4(j+c)(j+c+\alpha)(j+c+\beta)(j+c+\alpha+\beta)}{(2(j+c)+\alpha+\beta)^2(2(j+c)+\alpha+\beta+1)(2(j+c)+\alpha+\beta-1)}, \\
\text{Laguerre: } h_n^{(c)} &= \prod_{j=1}^{n}(j+c)(j+c+\alpha) = \frac{\Gamma(n+c+1)}{\Gamma(c+1)} \cdot \frac{\Gamma(n+c+\alpha+1)}{\Gamma(c+\alpha+1)}, \\
\text{Hermite: } h_n^{(c)} &= \prod_{j=1}^{n} \frac{j+c}{2} = \frac{\Gamma(n+c+1)}{2^n \, \Gamma(c+1)}.
\end{aligned} \tag{145}$$

For $c=1$, $\alpha=0$ the Laguerre product is $\Gamma(n+2)^2=\big((n+1)!\big)^2$, which gives $h_1^{(1)}=4$, $h_2^{(1)}=36$, $h_3^{(1)}=576$, in agreement with the numerical check below.

7.3 The associated weights

The mass distributions of the tails are related to one another by a formula that is the heart of the associated theory.

Proposition 23 (Associated weights)**.** *Let ν_c have density $\nu_c'(x)$ on an interval of the real line. Then*

$$\nu_c'(x)=\frac{\nu_{c-1}'(x)}{a_c^2\,\lvert F_{c-1}(x+\mathrm{i}0)\rvert^2},\qquad c\geq 1, \tag{146}$$

and, iterating,

$$\nu_c'(x)=\frac{\nu_0'(x)}{\prod_{j=1}^{c}a_j^2\,\lvert F_{j-1}(x+\mathrm{i}0)\rvert^2}. \tag{147}$$

Proof. From (134), $1/F_{c-1}(z)=z-b_{c-1}-a_c^2F_c(z)$. Take imaginary parts on the real axis from above. On the one hand

$$\operatorname{Im}\frac{1}{F_{c-1}}=-\frac{\operatorname{Im}F_{c-1}}{\lvert F_{c-1}\rvert^2}=\frac{\pi\nu_{c-1}'}{\lvert F_{c-1}\rvert^2}$$

by (132); on the other hand

$$\operatorname{Im}\frac{1}{F_{c-1}}=-a_c^2\operatorname{Im}F_c=\pi a_c^2\,\nu_c'.$$

Equating the two gives (146); (147) is its iteration. □

Corollary 24. *For every $c\geq 0$ the right-hand side of (147), integrated over the interval of the original weight, has total mass 1.*

Proof. By Proposition 23 it is the density of the spectral mass distribution ν_c, whose total mass is 1 by the construction of §6 applied to the finite sections of $J^{(c)}$. □

The corollary is what fixes the normalisation in concrete examples. The identity (146) is derived from the algebra of transforms alone, and its content is fixed only once the constant is known: the weight it produces must integrate to 1, with no adjustable factor. In each of the examples below this integral is checked by quadrature, and it is a sharp test of the algebra.

7.4 Three closed-form examples

For the three classical families the transforms of Proposition 18 are elementary, so the associated weights are too. In each case we list the transform F_0, the weight ν_1', the first associated polynomials and the norms, and we report what a double-precision check with the installed Maxima supports: the Legendre and Laguerre Gram matrices are checked to the accuracy stated below, and the Hermite weight is checked pointwise. No higher-precision claim is made.

(a) Associated Legendre. With $w = 1$ on $(-1, 1)$, so $h_0 = 2$, and $a_1^2 = \frac{1}{3}$,

$$F_0(z) = \frac{1}{2} \ln \frac{z+1}{z-1}, \qquad \nu_1'(x) = \frac{6}{\pi^2 + \ln^2 \frac{1+x}{1-x}} \quad (-1 < x < 1),$$

and zero outside. The first associated polynomials are

$$Q_0^{(1)} = 1, \quad Q_1^{(1)} = x, \quad Q_2^{(1)} = x^2 - \frac{4}{15}, \quad Q_3^{(1)} = x^3 - \frac{11}{21}x, \quad Q_4^{(1)} = x^4 - \frac{7}{9}x^2 + \frac{64}{945}, \tag{148}$$

with norms

$$h_0^{(1)} = 1, \quad h_1^{(1)} = \frac{4}{15}, \quad h_2^{(1)} = \frac{12}{175}, \quad h_3^{(1)} = \frac{64}{3675}.$$

The quadrature gives 1.000000, 0.266667, 0.068571, 0.017415 on the diagonal and off-diagonal entries below 10^{-9}.

Checking the weight by hand. On the cut, $\ln \frac{x+1}{x-1} = \ln \frac{1+x}{1-x} - \mathrm{i}\pi$ for $-1 < x < 1$ (the branch taken from above, as in §6), so $|F_0|^2 = \frac{1}{4}(\pi^2 + \ln^2 \frac{1+x}{1-x})$ and

$$\nu_1' = \frac{1/2}{(1/3) \cdot \frac{1}{4}(\pi^2 + \ln^2)} = \frac{6}{\pi^2 + \ln^2},$$

because $\nu_0' = w/h_0 = \frac{1}{2}$. The mass is

$$\int_{-1}^{1} \frac{6\,\mathrm{d}x}{\pi^2 + \ln^2 \frac{1+x}{1-x}} = 1, \tag{149}$$

which the quadrature confirms to twelve digits.

(b) Associated Laguerre at $\alpha = 0$. With $w = e^{-x}$ on $(0, \infty)$, so $h_0 = 1$, $a_1^2 = 1$, and

$$F_0(z) = e^{-z}\big(\mathrm{Ei}(z) - \mathrm{i}\pi\big), \qquad \nu_1'(x) = \frac{e^x}{\mathrm{Ei}(x)^2 + \pi^2} \quad (x > 0).$$

The first associated polynomials are

$$Q_0^{(1)} = 1, \quad Q_1^{(1)} = x - 3, \quad Q_2^{(1)} = x^2 - 8x + 11, \quad Q_3^{(1)} = x^3 - 15x^2 + 58x - 50, \tag{150}$$

with norms $h_0^{(1)} = 1$, $h_1^{(1)} = 4$, $h_2^{(1)} = 36$, $h_3^{(1)} = 576$, that is $h_n^{(1)} = \big((n+1)!\big)^2$ at $\alpha = 0$, in agreement with (145). The Gram matrix is diagonal to five digits, and the off-diagonal entries are below 10^{-5}.

Remark 25. *The polynomial $Q_2^{(1)} = x^2 - 8x + 11$ is a useful warning: its constant term comes from the recurrence. The tail coefficients are $b_0^{\mathrm{tail}} = b_1 = 3$, $b_1^{\mathrm{tail}} = b_2 = 5$ and $a_1^{\mathrm{tail},2} = a_2^2 = 4$, so $Q_2^{(1)} = (x-5)(x-3) - 4 = x^2 - 8x + 11$. Dropping the constant term $15 - 4 = 11$ (as one does when only the leading coefficients are tracked) gives a polynomial that fails orthogonality by a visible amount.*

(c) Associated Hermite. With $w = e^{-x^2}$ on $\mathbb{R}$, so $h_0 = \sqrt{\pi}$, $a_1^2 = \frac{1}{2}$, and

$$F_0(z) = -\mathrm{i}\sqrt{\pi}\, e^{-z^2}\operatorname{erfc}(-\mathrm{i}z), \qquad \nu_1'(x) = \frac{2\, e^{x^2}}{\pi^{3/2}\,|\operatorname{erfc}(-\mathrm{i}x)|^2}.$$

The first associated polynomials are

$$Q_0^{(1)} = 1, \quad Q_1^{(1)} = x, \quad Q_2^{(1)} = x^2 - 1, \quad Q_3^{(1)} = x^3 - \frac{5}{2}x, \tag{151}$$

with norms $1, 1, \frac{3}{2}, 3$. The weight formula was checked pointwise at $x = 1$ and $x = 2$ against direct quadrature of $|F_0(x + \mathrm{i}0)|^{-2}$; a full Gram matrix is not quoted because $\operatorname{erfc}(-\mathrm{i}x)$ loses accuracy for large $|x|$ in double precision. Note that $Q_2^{(1)} = x^2 - 1$ while the monic Hermite polynomial of degree two is $x^2 - \frac{1}{2}$; the tail uses $a_2^2 = 1$, not $a_2^2 = \frac{1}{2}$. This is the cleanest illustration of the difference between a family and its associate.

7.5 The associated families are not classical

It is natural to ask whether the associated family is again one of the classical families with different parameters. It is not, and the reason can be seen coefficient by coefficient.

Proposition 26. *For $c = 1$ none of the three associated families is Jacobi, Laguerre or Hermite.*

Proof. Legendre. The associated family has $b_n^{(1)} = b_{n+1} = 0$ for all n, so if it were Jacobi the two parameters would have to be equal, $\alpha = \beta$. Matching $a_1^{(1)2} = \frac{4}{15}$ with $a_1^2(\gamma, \gamma) = \dfrac{1}{3 + 2\gamma}$ gives $\gamma = \frac{3}{8}$. Then the Jacobi formula would give

$$a_2^2(\tfrac{3}{8}, \tfrac{3}{8}) = \frac{88}{345} = 0.25507\ldots,$$

whereas the associated family has $a_2^{(1)2} = \frac{9}{35} = 0.25714\ldots$; the two differ, so no choice of parameters works.

Laguerre. For the associated family $b_n^{(1)} = 2n + \alpha + 3$, which is the Laguerre coefficient with α replaced by $\alpha + 2$; but then the off-diagonal coefficient would have to be $a_n^2(\alpha + 2) = n(n + \alpha + 2)$, whereas the associated family has $a_n^{(1)2} = (n + 1)(n + \alpha + 1)$. At $n = 1$ these are $2\alpha + 4$ and $\alpha + 3$, which differ for every $\alpha \neq -1$.

Hermite. Both families have $b_n = 0$, but $a_n^{(1)2} = (n + 1)/2$ whereas $a_n^2 = n/2$; the sequences differ already at $n = 1$. □

The failure is not a defect: it is the content of Bochner's theorem. The associated family is orthogonal and satisfies a three-term recurrence, but its differential equation is no longer of second order with $\deg \sigma \leq 2$, $\deg \tau \leq 1$ and n-independent coefficients, so the theorem does not apply.

7.6 General c, continued fractions, and the TRA

For c an integer, the associated family is the family of the tail matrix $J^{(c)}$, and the whole picture of §6 applies with $b_n \to b_{n+c}$, $a_n^2 \to a_{n+c}^2$. The transforms nest as in (134), the weights iterate as in

(147), and the norms multiply as in (144). For non-integer c the same formulas define the family by continuation in the index; the coefficients b_{n+c}, a^2_{n+c} remain positive for c in the parameter range, so the continued fraction still represents a positive weight. This is the classical theory of associated polynomials (Geronimus; see [4, 5] for systematic treatments, and [8] for the explicit hypergeometric formulas in the Jacobi case).

There is a second way to see the construction, and it explains why the associated families matter beyond this note. The tridiagonal representation approach to exactly solvable problems in quantum mechanics [9] starts from a Jacobi matrix: the Hamiltonian (the energy operator of the problem) is written in a basis in which it is tridiagonal, and then the orthogonal polynomials of that matrix are the wavefunctions of the bound states. Changing the gauge, or the asymptotic parameter, of the problem changes the recurrence coefficients, and the simplest such change is exactly the index shift of (143). So the associated case computed here—shift the recurrence, keep the closed form of the weight—is the mechanism behind that approach, in its simplest setting, where the shifted recurrence is the tail of a classical one and the resulting weight can still be written down explicitly.

8 Zeros

8.1 The classical zeros

Since $Q_n = \det(xI - J_n)$, the zeros of Q_n are the eigenvalues of the $n \times n$ leading section of J. They are real, simple, lie in the interval of the weight—$[-1, 1]$ for Jacobi, $[0, \infty)$ for Laguerre, and the whole line for Hermite—and interlace with the zeros of Q_{n-1}. For Legendre this gives the familiar pattern $-\sqrt{\frac{3}{5}}, 0, \sqrt{\frac{3}{5}}$ for Q_3; for Laguerre the zeros are positive because the weight lives on $(0, \infty)$; for Hermite they are symmetric about the origin.

8.2 The associated zeros

The same statement holds verbatim for the associated family, with J replaced by the tail $J^{(c)}$; in particular the zeros of $Q^{(c)}_n$ are real and simple and interlace with those of $Q^{(c)}_{n-1}$. Because the densities of §7 are positive exactly where the original weight is positive, the associated zeros lie in the same interval: inside $(-1, 1)$ for Legendre, on $(0, \infty)$ for Laguerre, and on $\mathbb{R}$ for Hermite. The following table lists the zeros of the first few associated polynomials.

family	polynomial	zeros
Legendre	$x^2 - \frac{4}{15}$	± 0.5164
	$x^3 - \frac{11}{21}x$	$0,\ \pm 0.7237$
	$x^4 - \frac{7}{9}x^2 + \frac{64}{945}$	$\pm 0.3161,\ \pm 0.8233$
	$x^5 - \frac{34}{33}x^3 + \frac{1}{5}x$	$0,\ \pm 0.5094,\ \pm 0.8780$
Laguerre $\alpha = 0$	$x - 3$	3
	$x^2 - 8x + 11$	$1.7639,\ 6.2361$
	$x^3 - 15x^2 + 58x - 50$	$1.2104,\ 4.3990,\ 9.3906$
	$x^4 - 24x^3 + 177x^2 - 444x + 274$	$0.9052,\ 3.3898,\ 7.0645,\ 12.6405$
Hermite	x	0
	$x^2 - 1$	± 1
	$x^3 - \frac{5}{2}x$	$0,\ \pm 1.5811$
	$x^4 - \frac{9}{2}x^2 + 2$	$\pm 0.7071,\ \pm 2$

Table 4: Zeros of the first associated polynomials of order $c = 1$.

Two features are worth noticing. First, the roots are not the classical roots shifted: the associated Legendre zeros ± 0.7237 differ from the Legendre zeros $\pm\sqrt{3/5} = \pm 0.7746$, and all the associated zeros stay strictly inside the interval, as they must, because the weight lives on the same interval. Second, the interlacing pattern survives: the zeros of $x^4 - \frac{7}{9}x^2 + \frac{64}{945}$ at ± 0.3161 and ± 0.8233 interlace with $0, \pm 0.7237$.

8.3 Numerical checks collected

For the record, the numerical statements of this note were checked as follows.

1. The recurrences (49), (58), (64), the structure relations (68)–(70), and the differential equations (47), (57), (63) were checked for $n = 1, \dots, 5$ (Jacobi and Hermite symbolically in Maxima 5.49; Laguerre in 30-digit arithmetic).

2. The norm identity (72) was checked for $n \le 6$ in all three families; the residuals are at the level of the working precision.

3. The transforms (136)–(138) agree with quadrature to fourteen digits at $z = 0.4 + 0.7\mathrm{i}$.

4. The convergent identity (142) has residual below 3×10^{-15} for $n \le 4$ (Hermite, double precision).

5. The Legendre and Laguerre Gram matrices of the associated weights are diagonal to the accuracy stated in §7, and their diagonal entries equal $\prod_{j=1}^{n} a_{j+1}^2$; the associated Hermite weight was checked pointwise.

6. The limit tables of §5 show the expected rates $1/\beta$ and $\alpha^{-1/2}$.

7. The physics identities of §4 were checked symbolically in Maxima 5.49, with every residual identically zero: the generating function (88) against the Rodrigues formula for $\ell \le 6$; the Jacobi identity (96) for $\ell \le 6$; the reduction (97) of the associated Legendre equation for $m \le 4$ and $n \le 4$; the Gegenbauer relation (103) and the Gegenbauer equation for $\ell \le 6$ and $\lambda \in \{\frac{1}{2}, 1, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}\}$; the harmonicity of $r^{\ell} P_{\ell}(\cos\theta)$ and of $r^{-\ell-1} P_{\ell}(\cos\theta)$ for $\ell \le 5$; the hydrogenic radial functions (112) for $n \le 4$; the radial functions of the three-dimensional oscillator for

$n_r \le 3$ and $\ell \le 3$; and the oscillator wave functions (116) for $n \le 4$. The numerical statement $kc_k \to C \neq 0$ in the proof of Lemma 11 was computed in 30-digit arithmetic.

9 Common confusions

1. *"Classical" is not a synonym for "well known".* It means satisfying Definition 5. Bochner's theorem leaves exactly four families, and the associated families are not among them (Proposition 26).

2. *Monic versus standard.* The identity $h_n = a_n^2 h_{n-1}$ is stated for monic norms. In the standard normalisation it becomes (73), which contains the ratio k_n/k_{n-1}; forgetting it is the commonest source of factor errors in this subject.

3. *The J-fraction computes the probability transform.* The continued fraction (133) gives $\int \mathrm{d}\nu/(z-t)$ with $\nu = w\,\mathrm{d}x/h_0$, not $\int w\,\mathrm{d}x/(z-t)$; the two differ by the factor h_0, which for Hermite is $\sqrt{\pi}$. This is the error that is easiest to make in this subject, and it is the reason (135) is displayed separately.

4. *"Associated Legendre" is overloaded.* In physics the name usually refers to P_l^m, the derivative of the Legendre polynomial; in this note it refers to the recurrence-shifted family (143). The two objects are different.

5. *The parameter shift in (5) is in the parameters, not in the argument.* $Q_n'(x;\alpha) = nQ_{n-1}(x;\alpha+1)$ compares two different families at the same x; it is not a statement about $Q_n'(x+1)$.

6. *Bessel polynomials are classical but not orthogonal on the real line.* They satisfy Bochner's equation with $\sigma = x^2$, but the Pearson weight $x^{a-2}e^{-b/x}$ does not have finite moments on $(0,\infty)$ and is not positive on any interval containing negative x. Orthogonality with respect to a positive weight is an extra requirement that removes them.

7. *The associated family is not a parameter shift of the classical family.* $Q_n^{(1)}$ for Laguerre is not $Q_n^{(\alpha+2)}$ in the Laguerre family: the diagonal coefficients would agree but the off-diagonal ones would not, and the weights are different.

8. *The orthogonality weight is part of the coordinate system.* The Legendre polynomials are orthogonal with weight 1 in $x = \cos\theta$, which is the statement $\mathrm{d}\Omega = \mathrm{d}x\,\mathrm{d}\varphi$; in d dimensions the same computation gives the Jacobi weight $(1-x^2)^{(d-3)/2}$ (§4). The weight is the Jacobian of the coordinate change, not a normalisation chosen for convenience, and changing variables changes both the argument of the polynomials and the weight.

9. *The hydrogen radial functions are not Laguerre polynomials.* $R_{n\ell}(r)$ is $\rho^\ell e^{-\rho/2}$ times a Laguerre polynomial in $\rho = 2\kappa r$, and κ depends on n; so the orthogonality $\int R_{n\ell}R_{n'\ell}r^2\,\mathrm{d}r = 0$ is the Sturm–Liouville orthogonality of the radial equation, not the Laguerre orthogonality, which is an integral in a fixed variable ρ with weight $\rho^{2\ell+1}e^{-\rho}$ (§4.5).

10 Exercises

1. Prove (71) for Hermite and Laguerre from (70) and (69).

2. Show that the Hermite polynomials of even (odd) degree are even (odd) functions, and deduce $b_n = 0$ from the recurrence.

3. Compute Q_2 and Q_3 for Legendre from (56) and verify $a_n^2 = n^2/(4n^2-1)$ at $n = 1, 2, 3$.

4. Prove the norm identity (72) and deduce $h_n = h_0 \prod_{j=1}^{n} a_j^2$.

5. Verify the Pearson equation (34) for the three weights of Table 1, and derive the weights from σ and τ.

6. Derive the leading coefficient (48) from the Rodrigues formula (42).

7. Prove the Jacobi–Laguerre limit (122) for $n = 1, 2$ by direct expansion in $1/\beta$ (the case $n = 1$ works for general α).

8. Show that the continued fraction (133) converges to F_0^w/h_0 rather than to F_0^w, by expanding both sides in powers of $1/z$.

9. Derive the associated Laguerre weight of §7(b) from (146) and check that it integrates to 1.

10. Compute the first four associated Legendre polynomials from (143) and verify the norms $1, \frac{4}{15}, \frac{12}{175}, \frac{64}{3675}$.

11. Prove that the associated Legendre family is not a Jacobi family by the coefficient argument of Proposition 26.

12. Verify the convergent identity (142) for $n = 1$ for Hermite, using (134).

13. Show that $Q_k(J)e_0 = a_1 a_2 \cdots a_k \, e_k$ and deduce the moment identity (127) with $\nu = w\,\mathrm{d}x/h_0$ from it.

14. Compute the second convergent q_2/Q_2 for Legendre, expand it in powers of $1/z$, and check that it agrees with $\frac{1}{2}\ln\frac{z+1}{z-1}$ through the z^{-3} term while differing first in the coefficient of z^{-5}, as Proposition 20 predicts.

15. A conducting sphere of radius a is held at the potential $V_0 \cos^3\theta$. Expand $\cos^3\theta$ in Legendre polynomials, find the potential inside and outside the sphere, and identify the multipoles of each.

16. Compute P_2^1, P_2^2 and P_3^2 from the derivative formula (95), and verify the Jacobi identity (96) for $(\ell, m) = (2,1)$, $(2,2)$ and $(3,2)$ using $P_1^{(\alpha,\beta)} = \frac{1}{2}[(\alpha-\beta) + (\alpha+\beta+2)x]$.

17. Compute Y_1^0 and $Y_1^{\pm 1}$ from (98), and check that $\sum_m |Y_1^m|^2 = 3/(4\pi)$. Show that this is (101) at $\ell = 1$ and $\gamma = 0$.

11 Solutions

1. Hermite: $Q_n = 2^{-n}H_n$, so $Q_n' = 2^{-n}\cdot 2nH_{n-1} = n\cdot 2^{-(n-1)}H_{n-1} = nQ_{n-1}$. Laguerre: $Q_n = (-1)^n n! L_n^{(\alpha)}$, so $Q_n' = (-1)^n n!\cdot(-1)L_{n-1}^{(\alpha+1)} = (-1)^{n-1} n n! L_{n-1}^{(\alpha+1)} = nQ_{n-1}^{(\alpha+1)}$.

2. e^{-x^2} is even, so every derivative $\frac{\mathrm{d}^n}{\mathrm{d}x^n}e^{-x^2}$ has the parity $(-1)^n$; multiplying by $(-1)^n$ gives $H_n(-x) = (-1)^n H_n(x)$. In the recurrence $H_{n+1} = 2xH_n - 2nH_{n-1}$ the two terms on the right have the same parity as H_{n+1} for every n, so comparing the coefficient of x^{n+1} on both sides gives $b_n = 0$ (the coefficient of x^n in H_{n+1} vanishes).

3. $Q_2 = xQ_1 - a_1^2 = x^2 - \frac{1}{3}$ and $Q_3 = xQ_2 - a_2^2 Q_1 = x^3 - \frac{1}{3}x - \frac{4}{15}x = x^3 - \frac{3}{5}x$. With $a_n^2 = n^2/(4n^2-1)$: $a_1^2 = \frac{1}{3}$, $a_2^2 = \frac{4}{15}$, $a_3^2 = \frac{9}{35}$.

4. This is Proposition 10; iterating gives $h_n = h_0 a_1^2 a_2^2 \cdots a_n^2$.

5. Jacobi: with $w = (1-x)^\alpha(1+x)^\beta$, $w'/w = -\alpha/(1-x) + \beta/(1+x)$; multiplying by $1-x^2$ gives $\beta - \alpha - (\alpha+\beta)x = \tau + 2x$, i.e. $(1-x^2)w' = (\tau+2x)w = (\sigma w)'$ since $\sigma' = -2x$. Laguerre: $xw' = (\alpha - x)w = (\tau - 1)w$, and $(\sigma w)' = w + xw'$, so $(\sigma w)' = \tau w$. Hermite: $w' = -2xw = \tau w$ and $\sigma' = 0$.

6. Apply $\frac{\mathrm{d}^n}{\mathrm{d}x^n}$ to $(1-x)^{n+\alpha}(1+x)^{n+\beta}$: the top term is $(-1)^n n!\,(1-x)^\alpha(1+x)^\beta x^n + \dots$; multiply by $(-1)^n/(2^n n!)(1-x)^{-\alpha}(1+x)^{-\beta}$ gives $x^n \cdot n!/(2^n n!)$ times the coefficient, which is the leading coefficient of $P_n^{(\alpha,\beta)}$, namely $(n+\alpha+\beta+1)_n/(2^n n!)$. (Equivalently, differentiate the product n times and keep the term in which all n derivatives hit the x^n factor.)

7. For $n = 1$,

$$\beta Q_1^{(\alpha,\beta)}\Big(1 - \frac{2z}{\beta}\Big) = \beta - 2z + \frac{\beta(\alpha-\beta)}{\beta+\alpha+2} = 2\alpha + 2 - 2z + O(\beta^{-1}),$$

using $\beta(\alpha-\beta)/(\beta+\alpha+2) = \alpha - \beta + \alpha + 2 + O(\beta^{-1})$, and $2\alpha + 2 - 2z = -2Q_1^L(z;\alpha)$. For $n = 2$ the same computation with $Q_2 = (x-b_1)(x-b_0) - a_1^2$ gives $\beta^2 Q_2(1 - 2z/\beta) \to 4Q_2^L(z;\alpha)$, consistent with $(-2)^2$.

8. Expand the continued fraction in powers of $1/z$: $F_0 = 1/z + b_0/z^2 + (b_0^2 + a_1^2)/z^3 + \cdots$, so the coefficient of $1/z$ is 1, which is the mass of ν. The transform of $w\,\mathrm{d}x$ has coefficient h_0/z; hence the continued fraction is F_0^w/h_0.

9. With $\nu_0' = e^{-x}$, $a_1^2 = 1$ and $F_0 = e^{-z}(\mathrm{Ei}(z) - \mathrm{i}\pi)$,

$$\nu_1' = \frac{e^{-x}}{1 \cdot |e^{-x}(\mathrm{Ei}(x) - \mathrm{i}\pi)|^2} = \frac{e^{-x}}{e^{-2x}(\mathrm{Ei}(x)^2 + \pi^2)} = \frac{e^x}{\mathrm{Ei}(x)^2 + \pi^2}.$$

Its mass is one by Corollary 24; the quadrature confirms it.

10. $Q_0 = 1$, $Q_1 = x$, $Q_2 = x^2 - a_2^2 = x^2 - \frac{4}{15}$, $Q_3 = xQ_2 - a_3^2 Q_1 = x^3 - (\frac{4}{15} + \frac{9}{35})x = x^3 - \frac{11}{21}x$, $Q_4 = xQ_3 - a_4^2 Q_2 = x^4 - \frac{11}{21}x^2 - \frac{16}{63}(x^2 - \frac{4}{15}) = x^4 - \frac{7}{9}x^2 + \frac{64}{945}$. The norms follow from (144) with $h_0^{(1)} = 1$: $\frac{4}{15}$, $\frac{4}{15}\cdot\frac{9}{35} = \frac{12}{175}$, $\frac{12}{175}\cdot\frac{16}{63} = \frac{64}{3675}$.

11. This is the first paragraph of the proof of Proposition 26: $b_n^{(1)} = 0$ forces $\alpha = \beta$; a_1^2 then forces $\gamma = \frac{3}{8}$; but $a_2^2(\frac{3}{8}, \frac{3}{8}) = \frac{88}{345} \neq \frac{9}{35} = a_2^{(1)2}$.

12. For $n = 1$, (142) reads $Q_1 F_0 - q_1 = a_1^2 F_0 F_1$. But $F_0 = 1/(z - b_0 - a_1^2 F_1)$, so $F_0(z - b_0 - a_1^2 F_1) = 1$; since $Q_1 = z - b_0$ and $q_1 = 1$, this is $Q_1 F_0 - 1 = a_1^2 F_0 F_1$, as claimed.

13. Induct on k. For $k = 0$ the statement is $e_0 = e_0$. Assume $Q_k(J)e_0 = a_1 \cdots a_k e_k$. Then, using $Je_k = a_k e_{k-1} + b_k e_k + a_{k+1} e_{k+1}$,

$$Q_{k+1}(J)e_0 = \big(J - b_k\big)Q_k(J)e_0 - a_k^2 Q_{k-1}(J)e_0 = a_1 \cdots a_k\big(a_k e_{k-1} + a_{k+1}e_{k+1}\big) - a_k^2 a_1 \cdots a_{k-1} e_{k-1},$$

and the e_{k-1} terms cancel, leaving $a_1 \cdots a_{k+1} e_{k+1}$. Now expand $t^k = Q_k(t) + c_{k-1}Q_{k-1}(t) + \cdots + c_0$; the e_0 entry of $J^k e_0$ is the e_0 entry of the corresponding combination of $Q_j(J)e_0$, and since $Q_j(J)e_0$

has no e_0 component for $j \geq 1$, that entry is c_0. Integrating the same expansion, the terms with $j \geq 1$ drop out by orthogonality and the Q_0 term gives $c_0 h_0/h_0 = c_0$. Hence $(e_0, J^k e_0) = \frac{1}{h_0}\int t^k w\,\mathrm{d}t$.

14. For Legendre, $b_0 = b_1 = 0$, $a_1^2 = \frac{1}{3}$, so $q_2 = (z - b_1)q_1 - a_1^2 q_0 = z$ and $Q_2 = z^2 - \frac{1}{3}$. Thus

$$\frac{q_2}{Q_2} = \frac{z}{z^2 - \frac{1}{3}} = \frac{1}{z}\Big(1 + \frac{1}{3z^2} + \frac{1}{9z^4} + \cdots\Big) = \frac{1}{z} + \frac{1}{3z^3} + \frac{1}{9z^5} + \cdots.$$

On the other hand $\frac{1}{2}\ln\frac{z+1}{z-1} = \frac{1}{2}\ln\frac{1+1/z}{1-1/z} = \frac{1}{z} + \frac{1}{3z^3} + \frac{1}{5z^5} + \cdots$. The two agree through z^{-3} and differ first at z^{-5}, which is the order $z^{-2n-1} = z^{-5}$ predicted by (142) for $n = 2$.

15. $P_1 = x$ and $P_3 = \frac{1}{2}(5x^3 - 3x)$ give $x^3 = \frac{3}{5}P_1(x) + \frac{2}{5}P_3(x)$, hence

$$\cos^3\theta = \tfrac{3}{5}P_1(\cos\theta) + \tfrac{2}{5}P_3(\cos\theta).$$

The boundary data therefore contain only $\ell = 1$ and $\ell = 3$, and the comparison of coefficients of Example 12 gives

$$V_{\text{in}} = V_0\Big[\tfrac{3}{5}\Big(\frac{r}{a}\Big)P_1(\cos\theta) + \tfrac{2}{5}\Big(\frac{r}{a}\Big)^3 P_3(\cos\theta)\Big],$$
$$V_{\text{out}} = V_0\Big[\tfrac{3}{5}\Big(\frac{a}{r}\Big)^2 P_1(\cos\theta) + \tfrac{2}{5}\Big(\frac{a}{r}\Big)^4 P_3(\cos\theta)\Big].$$

The outside potential is a dipole plus an octupole. There is no monopole because $\int_{-1}^{1}\cos^3\theta\,\mathrm{d}x = 0$, and no quadrupole because $\cos^3\theta$ is odd while P_2 is even; the two coefficients are the projections (87).

16. $P_2 = \frac{1}{2}(3x^2 - 1)$ gives $P_2' = 3x$ and $P_2'' = 3$, so $P_2^1 = 3x(1 - x^2)^{1/2}$ and $P_2^2 = 3(1 - x^2)$. $P_3 = \frac{1}{2}(5x^3 - 3x)$ gives $P_3' = \frac{1}{2}(15x^2 - 3)$ and $P_3'' = 15x$, so $P_3^2 = 15x(1 - x^2)$. On the other side, $P_1^{(1,1)} = 2x$ and $P_1^{(2,2)} = 3x$, and (96) gives

$$(\ell, m) = (2, 1): \quad \frac{3!}{2 \cdot 2!}(1 - x^2)^{1/2} \cdot 2x = 3x(1 - x^2)^{1/2},$$
$$(\ell, m) = (2, 2): \quad \frac{4!}{4 \cdot 2!}(1 - x^2) \cdot 1 = 3(1 - x^2),$$
$$(\ell, m) = (3, 2): \quad \frac{5!}{4 \cdot 3!}(1 - x^2) \cdot 3x = 15x(1 - x^2),$$

which agree with the derivatives.

17. $P_1^1 = (1 - x^2)^{1/2} = \sin\theta$, and (98) gives

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta, \qquad Y_1^1 = \sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\varphi}, \qquad Y_1^{-1} = -\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{-i\varphi}.$$

Hence

$$\sum_{m=-1}^{1} |Y_1^m|^2 = \frac{3}{4\pi}\cos^2\theta + 2 \cdot \frac{3}{8\pi}\sin^2\theta = \frac{3}{4\pi}.$$

The addition theorem (101) at $\ell = 1$ reads, for $\gamma = 0$ (equal directions), $P_1(1) = 1 = \frac{4\pi}{3}\sum_m |Y_1^m|^2$, which is the same identity multiplied by $4\pi/3$.

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